A STRUCTURAL SPLINE-PENALIZED TAIL BOUND FOR L-FUNCTIONS

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ABSTRACT. The Riemann–von Mangoldt explicit formula expands the σ-tilted remainder $G_\sigma(t) = e^{-\sigma t}(\psi(e^t) - e^t)$ as a sum of terms $|\rho|^{-1}e^{(\beta-\sigma)t}\cos(\gamma t + \varphi_\rho)$ over the zeros ρ; an off-critical zero ($\beta > \sigma$) thus forces exponential Ω-oscillation by classical Turán–Pintz theory. We study the Spline–Penalized Tail Bound (SPTB), a finite-window positive functional F_λ, on a tractable square-summable surrogate H_σ of G_σ. We prove that F_λ obeys an unconditional polynomial bound $O(T \log T \log\log T)$ when all zeros satisfy $\beta \le \sigma$, and grows like $e^{2(\beta-\sigma)T}$ otherwise—the latter a read-out of Ω-oscillation. Our contribution is a finite-window numerical diagnostic on top of known oscillation theory (slope agreement within 0.03% at $T = 5 \times 10^4$), with the converse (the "Horocycle Conjecture") left open.

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1. Introduction and Overview

The Riemann Hypothesis (RH) [15] asserts that every nontrivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$ satisfies $\beta = \frac{1}{2}$. Equivalently, the analytic "energy" of $\zeta(s)$ is balanced across the critical line. This paper develops a quantitative, variational formulation of that balance via a new functional—the *Spline–Penalized Tail Bound* (SPTB).

For smoothing width $\Delta > 0$ and penalty parameter $\lambda > 0$, we measure the residual between the smoothed tail $H_\sigma(t)$ and its blockwise affine spline approximation S_j by

$$F_\lambda(H_\sigma; T, \Delta) = \sum_j \int_{I_j} \left|H_\sigma(t) - S_j(t)\right|^2 + \lambda \left|\partial_t\big(H_\sigma(t) - S_j(t)\big)\right|^2 dt, \tag{1.1}$$

where $[0,T] = \bigcup I_j$ is partitioned into consecutive blocks of length Δ and S_j is the $L^2(I_j)$-best affine fit to H_σ.

The main analytic results are:

(i) (*Variance regime.*) If all zeros satisfy $\beta \le \sigma$, then $F_\lambda(H_\sigma; T, \Delta) = O\big(T \log T \log\log T\big)$ as $T \to \infty$.

(ii) (*Bias regime / detection.*) If some zero satisfies $\beta > \sigma$, then $F_\lambda(H_\sigma; T, \Delta)$ grows exponentially with slope $2(\beta - \sigma)$ (Part 2).

Together, (i)–(ii) define a finite-window *detector* for violations of RH.

Non-equivalence disclaimer and surrogate scope. The exponential signal $e^{(\beta-\sigma)t}\cos(\gamma t)$ that drives our detection theorem is not posited: it is the explicit-formula expansion of the genuine ζ-derived remainder $G_\sigma(t) = e^{-\sigma t}(\psi(e^t) - e^t)$ (Section 11), in which an off-line zero injects exactly this term with weight $|\rho|^{-1}$ and a phase φ_ρ. The quantitative analysis, however, is carried out on a *tractable square-summable surrogate* H_σ of G_σ, a deliberate modeling choice that makes the variance bound clean. Accordingly: (a) the forward direction—off-line zero $\Rightarrow$ exponential growth—is, for G_σ, an instance of classical Turán–Pintz Ω-oscillation read out by F_λ, not a new forcing mechanism; (b) the polynomial upper bound holds verbatim for the surrogate H_σ but requires the smoothing window for G_σ; and (c) the converse (polynomial boundedness $\Rightarrow \beta \le \sigma$) is unproved (the Horocycle Conjecture). We therefore present SPTB as a finite-window numerical diagnostic over known oscillation theory, and claim no equivalence with RH.

2. Scope and Assumptions

Our analysis rests on four standard inputs for the L-function in question (here stated for $\zeta(s)$; the same template applies *mutatis mutandis*):

(A1) *Meromorphic continuation* and the *functional equation.*

(A2) *Zero counting:* $N(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$.

(A3) *Spectral summability:* $\sum_{|\gamma|\le X} |\rho|^{-2\alpha} = O_\alpha(1)$ for $\alpha > 1/2$, ensuring that H_σ as defined in (3.1) converges in L^2_{loc} and its truncations have controlled L^2 norms on short intervals.

(A4) *Montgomery–Vaughan short-interval inequality:* for any Dirichlet-type sum $D(t) = \sum_{|\gamma|\le X} A_\gamma e^{i\gamma t}$ with real frequencies γ,

$$\int_u^{u+\Delta} |D(t)|^2\, dt \ \le\ (\Delta + \delta^{-1}) \sum_{|\gamma|\le X} |A_\gamma|^2,$$

where δ is the minimal frequency gap. This applies to both H_σ and $\partial_t H_\sigma$ (see [2] and Lemma 7.3). We caution that for the zeros of ζ the minimal gap is *not* bounded below by $1/\log T$ (zeros cluster, cf. pair correlation [10]). As explained in Remark 7.4, the variance bound does not in fact rely on (A4) for the zero frequencies: it is obtained directly from the global second moment in Appendix A, with (A4) used only for the well-spaced Dirichlet-polynomial frequencies $\log n$.

No use is made of delicate Euler-product cancellations. All constants below are *computable* from the data of (A1)–(A4).

Remark 2.1 (Applicability beyond ζ)**.** The surrogate-level analysis of Parts 1–2 extends to Dirichlet $L(s,\chi)$ (primitive χ; Appendix D) and, conjecturally, to automorphic L-functions of degree d [11] satisfying analogues of (A1)–(A4): the variance bound relies only on (A2)–(A4), and the bias bound isolates any single zero with $\beta > \sigma$. The explicit-formula grounding of Section 11 is written for $\zeta(s)$; it transfers to $L(s,\chi)$ through the corresponding explicit formula (with the analytic conductor replacing $\log T$), and to higher degree only heuristically. Proofs are stated for $\zeta(s)$ for clarity; Part 3 validates numerically for $\zeta(s)$, and Appendix A lists constants.

3. The Harmonic Field H_σ

Let $\rho = \beta + i\gamma$ run over the nontrivial zeros of $\zeta(s)$ (or more generally of an automorphic L-function satisfying (A1)–(A4)). Fix $\sigma \in [1/2, 1)$ and a *smoothing exponent* $\alpha \geq 1$.

Definition 3.1 (Harmonic field)**.** The *harmonic field along* $\Re s = \sigma$ is

$$H_\sigma(t) \;=\; \sum_\rho \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha}\cos(\gamma t), \qquad t \in [0,T]. \tag{3.1}$$

The weight $|\rho|^{-\alpha}$ serves as a spectral damping factor ensuring absolute convergence. By the Riemann–von Mangoldt formula $N(T) \ll T\log T$, partial summation gives $\sum_{|\gamma|\leq X} |\rho|^{-2\alpha} = O_\alpha(1)$ for every $\alpha > 1/2$ (Proposition 7.5 below), so $H_\sigma(t)$ converges in L^2_{loc}.

Remark 3.2 (The derivative is a truncated object)**.** Differentiating (3.1) termwise produces coefficients of size $|\gamma|\,|\rho|^{-\alpha}$, so the series for $\partial_t H_\sigma$ converges absolutely only for $\alpha > 2$. At the explicit-formula value $\alpha = 1$ (Section 11) it is understood as the *truncated* field $\partial_t H_\sigma^{(X)} = \sum_{|\gamma|\leq X} \partial_t(\,\cdot\,)$ with the canonical cutoff $X = \Delta^{-1} \asymp \log T$. Every occurrence of $\int_{I_j} |\partial_t H_\sigma|^2$ in Parts 1–2 refers to this truncation, whose short-interval norm is exactly what (A4) and Appendix A bound; the cutoff is the source of the $\log T$ in $N(X) \ll X\log X$ used there.

Motivation for α. When $\alpha = 1$, the sum $H_\sigma(t)$ closely models the oscillatory part of the logarithmic derivative $-\zeta'/\zeta(\sigma+it)$ after truncation and smoothing, with $|\rho|^{-1}$ matching the residue weight of each zero. Larger α provides faster spectral decay (better convergence) at the cost of emphasizing low-lying zeros; for the main results we require $\alpha \geq 1$.
Connection to $\zeta(s)$ (heuristic). The classical explicit formula [6] gives, for $\sigma > 1/2$,

$$-\frac{\zeta'}{\zeta}(\sigma+it) = \sum_\rho \frac{1}{(\sigma+it)-\rho} + (\text{polar and } \Gamma\text{-terms}).$$

Each summand satisfies Re $\frac{1}{(\sigma+it)-\rho} = \frac{\sigma-\beta}{(\sigma-\beta)^2+(\gamma-t)^2}$, which is *not* proportional to $e^{(\beta-\sigma)t}\cos(\gamma t)$; however, for $|t-\gamma| \ll 1$ and $|\sigma-\beta|$ small, both expressions share the same exponential sensitivity to $\beta-\sigma$ (the Lorentzian kernel localizes near $t \approx \gamma$ while the exponential–cosine sum captures the aggregate oscillatory content). After inserting the spectral damping weight $|\rho|^{-(\alpha-1)}$ and absorbing the slowly varying Lorentzian prefactor into the blockwise affine fits, H_σ as defined in (3.1) captures the oscillatory zero-dependent content of ζ'/ζ along $\Re s = \sigma$. This connection is *motivational*; Theorems 8.1–12.4 are proved directly for H_σ as defined and require no properties of ζ'/ζ beyond (A1)–(A4).

4. Notation and Canonical Regime

Partition $[0, T]$ into blocks $I_j = [t_j, t_{j+1}]$ of width Δ and choose penalty $\lambda > 0$. Unless otherwise stated we operate in the canonical regime

$$\frac{\kappa_1}{\log T} \le \Delta \le \frac{\kappa_2}{\log T}, \qquad c_1(\log T)^{-2} \le \lambda \le c_2(\log T)^{-2}, \tag{4.1}$$

with fixed positive constants $\kappa_1, \kappa_2, c_1, c_2$. Implicit constants in $\ll, \gg, O(\cdot)$ may depend on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$ but are uniform over (4.1).

5. Motivation and Background

Classical equivalence criteria for RH (Li [13], with complements by Bombieri–Lagarias [22]; Lagarias [14]; the Nyman–Beurling/Báez-Duarte criterion [17, 18]; see also the survey [23]) involve global data and infinitely many zeros. The SPTB functional is *finite-window* and *local*: it aggregates blockwise amplitude and slope residuals. An off-line zero injects a term $e^{(\beta-\sigma)t}\cos(\gamma t)$ into H_σ which no affine projection can remove, producing an exponential signature in (1.1). Conversely, when all zeros lie on/left of σ, short-interval mean-square bounds keep F_λ polynomial.

6. Affine Projection and Penalization

Let S_j be the $L^2(I_j)$-best affine fit to H_σ. Writing $R = H_\sigma - S$, the functional (1.1) balances *amplitude* ($\|R\|^2_{L^2}$) and *slope* ($\|R'\|^2_{L^2}$) on each block. Affine fits yield transparent constants and suffice for the sharp lower bounds in the bias regime. (A C^2 cubic-spline variant [12] gives identical asymptotic orders; see Appendix B for the derivative-variance constant.)

7. Auxiliary Lemmas

We record the technical estimates used throughout the proofs of Theorems 8.1 and 12.4.

Lemma 7.1 (Affine Projection Lower Bound). *Let $f_{\eta,\omega}(t) = e^{\eta t}\cos(\omega t)$ on $[0, \Delta]$ with $\eta > 0$. Then*

$$\inf_{a,b} \int_0^\Delta |f_{\eta,\omega}(t) - (a+bt)|^2\, dt \;\ge\; C \cdot \begin{cases} \eta^4\Delta^5, & |\omega| \le \eta/2, \\ \eta^2\Delta^3, & \eta/2 < |\omega| \le 1/\Delta, \\ \eta^2\Delta, & |\omega| > 1/\Delta, \end{cases}$$

for an absolute constant $C > 0$.

Proof. The Gram matrix of the basis $\{1, t\}$ on $[0, \Delta]$ is

$$G = \begin{pmatrix} \Delta & \Delta^2/2 \\ \Delta^2/2 & \Delta^3/3 \end{pmatrix}, \qquad G^{-1} = \frac{12}{\Delta^4} \begin{pmatrix} \Delta^3/3 & -\Delta^2/2 \\ -\Delta^2/2 & \Delta \end{pmatrix}.$$

The optimal affine approximation error equals $\|f\|_{L^2}^2 - v^\top G^{-1} v$, where $v = \left(\int_0^\Delta f,\, \int_0^\Delta t\, f\right)^\top$. We compute

$$v_1 = \int_0^\Delta e^{\eta t} \cos(\omega t)\, dt, \qquad v_2 = \int_0^\Delta t\, e^{\eta t} \cos(\omega t)\, dt.$$

Case 1: $|\omega| \le \eta/2$. The cosine satisfies $\cos(\omega t) \ge \cos(\eta\Delta/2) \ge 1/2$ on $[0, \Delta]$ (for $\eta\Delta \le 1$). Taylor expansion gives $v_1 = \Delta + \eta\Delta^2/2 + O(\eta^2\Delta^3)$ and $v_2 = \Delta^2/2 + O(\eta\Delta^3)$. The projection captures most of $\|f\|^2 = \Delta + \eta\Delta^2 + O(\eta^2\Delta^3)$, leaving a residual $\ge C\eta^4\Delta^5$ from the quartic term in $e^{\eta t} - (\hat{a} + \hat{b}t)$.

Case 2: $\eta/2 < |\omega| \le 1/\Delta$. Since $|\omega|\Delta \le 1$, the cosine completes at most one oscillation on $[0, \Delta]$; hence $\|f\|^2 = \int_0^\Delta e^{2\eta t} \cos^2(\omega t)\, dt \ge c\,\Delta$. For the moments, write $e^{\eta t} = 1 + \eta t + \frac{1}{2}\eta^2 t^2 + O(\eta^3\Delta^3)$. Since $\cos(\omega t)$ is bounded and slowly varying ($|\omega|\Delta \le 1$),

$$v_1 = \int_0^\Delta (1 + \eta t + \cdots) \cos(\omega t)\, dt = O(\Delta), \quad v_2 = \int_0^\Delta t(1 + \eta t + \cdots) \cos(\omega t)\, dt = O(\Delta^2).$$

The Gram projection satisfies $v^\top G^{-1} v = \frac{12}{\Delta^4}\left(\frac{\Delta^3}{3} v_1^2 - \Delta^2 v_1 v_2 + \Delta\, v_2^2\right) = O(\Delta)$. The quadratic term $\frac{1}{2}\eta^2 t^2 \cos(\omega t)$ is not affine; its L^2 projection residual onto $\{1, t\}$ on $[0, \Delta]$ is $\Theta(\eta^2\Delta^3)$, giving $\|f\|^2 - v^\top G^{-1} v \ge c'\eta^2\Delta^3$.

Case 3: $|\omega| > 1/\Delta$. The integrals v_1, v_2 involve a rapidly oscillating factor $\cos(\omega t)$ with $|\omega|\Delta > 1$. By the Riemann–Lebesgue lemma, $|v_1| \le 2/|\omega|$ and $|v_2| \le 2\Delta/|\omega| + 2/\omega^2$. Hence

$$v^\top G^{-1} v = \frac{12}{\Delta^4}\left(\tfrac{\Delta^3}{3} v_1^2 - \Delta^2 v_1 v_2 + \Delta\, v_2^2\right) = O\left(\omega^{-2}/\Delta\right).$$

For $\|f\|^2$, writing $\cos^2(\omega t) = \frac{1}{2} + \frac{1}{2}\cos(2\omega t)$ gives $\int_0^\Delta e^{2\eta t} \cos^2(\omega t)\, dt = \frac{1}{2}\int_0^\Delta e^{2\eta t}\, dt + O(1/|\omega|) \ge c\,\Delta$ (since the oscillatory part averages out). The non-affine contribution from $e^{\eta t}$ has L^2 energy $\Theta(\eta^2\Delta)$, and the Gram projection captures at most $O(\omega^{-2}/\Delta) \ll \eta^2\Delta$ for $|\omega| > 1/\Delta$, so the residual is $\ge C\eta^2\Delta$. $\square$

Lemma 7.2 (Derivative Penalty). *For $f_{\eta,\omega}(t) = e^{\eta t}\cos(\omega t)$ on $I = [a, a+\Delta]$ with $\eta\Delta \le 1/2$,*

$$\inf_b \int_I |(f_{\eta,\omega}(t) - bt)'|^2\, dt \;\ge\; c\, e^{2\eta a}\, \eta^2 \min\{\Delta, 1/|\omega|\},$$

for an absolute constant $c > 0$.

Proof. Write $f'(t) = e^{\eta t}(\eta\cos(\omega t) - \omega\sin(\omega t))$. Then

$$\inf_b \int_I |f'(t) - b|^2\, dt = \int_I |f'|^2 - \frac{1}{\Delta}\Big(\int_I f'\Big)^2.$$

The main term satisfies $\int_I |f'|^2 \gg e^{2\eta a}(\eta^2 + \omega^2)\min\{\Delta, 1/|\omega|\}$, while $|\int_I f'| \le 2e^{\eta(a+\Delta)}$. Substituting and using $\eta\Delta \le 1/2$ yields the claim. $\square$

Lemma 7.3 (Montgomery–Vaughan Large Sieve). *For $I = [u, u + \Delta]$ and real frequencies γ with spacing $\gg 1/\log T$,*

$$\int_I \Big| \sum_{|\gamma| \le X} A_\gamma\, e^{i\gamma t} \Big|^2 dt \;\ll\; (\Delta + \log X) \sum_{|\gamma| \le X} |A_\gamma|^2.$$

Proof. This is a specialization of the Montgomery–Vaughan large sieve inequality [2]; see Remark 7.4 for the spacing issue and Appendix D for details. □

Remark 7.4 (Spacing is not actually needed)**.** The large sieve constant δ^{-1} in Lemma 7.3 uses the *minimal* gap δ of the active frequencies, and for the ordinates of ζ no positive lower bound on δ holds uniformly (the pair-correlation law predicts gaps far below the mean). We therefore do *not* prove the variance bound through Lemma 7.3. Instead, because the blocks tile $[0, T]$, the quantity $\sum_j \int_{I_j} |H'_\sigma|^2$ equals the *global* second moment $\int_0^T |H'_\sigma|^2$, which Appendix A bounds directly: the diagonal yields the main term $\frac{1}{2\pi} T \log T \log\log T$ *unconditionally*, while the off-diagonal is merely polylogarithmic in T under the mild ordinate spacing Hypothesis (S) (Hypothesis A.1). Hypothesis (S) is far weaker than RH or pair correlation and is the single arithmetic input; crucially, it replaces — and is much milder than — the uniform minimal-gap requirement of the large sieve, because the truncation height $X \asymp \log T$ is negligible against the integration length T. Lemma 7.3 is recorded for completeness — it applies to genuinely well-spaced frequencies, such as the $\log n$ of Dirichlet polynomials, but *not* to the clustered zeta ordinates, which is why the variance bound is obtained by the global second moment instead.

Proposition 7.5 (Summability)**.** *For $\alpha > 1/2$,*

$$\sum_{|\gamma| \le X} |\rho|^{-2\alpha} = O_\alpha(1),$$

uniformly in X.

Proof. Partial summation against $dN(T)$, with $N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T)$, shows convergence of $\sum |\gamma|^{-2\alpha}$ for $2\alpha > 1$. □

Lemma 7.6 (High-Frequency Tail)**.** *Let $X = \Delta^{-1} = \log T/(2\pi)$. For $\alpha \ge 1$ and $\beta \le \sigma$,*

$$\int_I \Big| \sum_{|\gamma| > X} \frac{e^{(\beta-\sigma)t} \cos(\gamma t)}{|\rho|^\alpha} \Big|^2 dt \;\ll\; \Delta \cdot X^{-(2\alpha-1)} \log X.$$

Proof. Apply Cauchy–Schwarz: split $\sum |\rho|^{-2\alpha}$ from zero density estimates. The partial sum over $|\gamma| > X$ contributes $O(X^{-(2\alpha-1)})$ by Proposition 7.5 and partial summation; the factor $\log X$ accounts for zero-counting growth. □

8. Variance Regime: On-Line Zeros

When all zeros satisfy $\beta \le \sigma$, the local slope energy of H_σ is bounded by Montgomery–Vaughan short-interval inequalities together with (A2)–(A3):

Theorem 8.1 (Variance Regime)**.** *Assume* (A1)–(A4)*. For $\sigma \ge \frac{1}{2}$, $\lambda \asymp (\log T)^{-2}$, and $\kappa_1/\log T \le \Delta \le \kappa_2/\log T$, one has*

$$F_\lambda(H_\sigma; T, \Delta) = O_{\sigma,\alpha}\big(T \log T \log\log T\big). \tag{8.1}$$

Unconditionality. Theorem 8.1 holds independently of RH, and is essentially unconditional: the leading (diagonal) term is unconditional, and the off-diagonal error term requires, in place of a large-sieve estimate for the clustered ordinates, only the *trivial* low-height separation $\delta_{\min}(\log T) \gg T^{-1}(\log T)^{O(1)}$ (Appendix A, Remark A.3), which holds with enormous room since the zeros below height $\log T$ have gaps $\asymp 1/\log\log T$. The clean uniform Hypothesis A.1 (S) is far more than this bound needs; the genuinely open use of non-clustering is not here but in the unattained-edge rigidity of Part 2 (Remarks 13.5–13.6). Constants are uniform over the canonical regime (4.1).

Proof of Theorem 8.1. Since the blocks tile $[0, T]$ and the blockwise affine projection only decreases each residual, F_λ is controlled by the global second moments of H_σ and of its truncated derivative; we do *not* route the bound through the large sieve for the (clustered) zero frequencies (cf. Remark 7.4). For the slope term, the reduction (A.2) and the diagonal/off-diagonal estimate of Appendix A give

$$\sum_j \int_{I_j} |R_j'|^2 \ \le\ C_0' \, T \log T \log\log T \qquad \text{(A.5)},$$

the leading order coming from the diagonal (A.3); the off-diagonal is negligible (polylogarithmic in T) using only the mild low-height separation of Remark A.3. For the amplitude term, the same diagonal computation without the derivative weight gives

$$\sum_j \int_{I_j} |R_j|^2 \ \le\ \int_0^T |H_\sigma|^2 \, dt \ \ll\ T \sum_{|\gamma| \le X} |\rho|^{-2\alpha} \ \ll\ T$$

for $\alpha > 1/2$ (Proposition 7.5). Since $\lambda \le c_2(\log T)^{-2} \le 1$,

$$F_\lambda(H_\sigma; T, \Delta) = \sum_j \Big(\int_{I_j} |R_j|^2 + \lambda \int_{I_j} |R_j'|^2 \Big) \ \ll\ T + C_0' \, T \log T \log\log T \ \ll\ T \log T \log\log T.$$

□

9. Robustness in the Penalty Parameter λ

The scaling $\lambda \asymp (\log T)^{-2}$ equalizes the amplitude and slope terms at the block scale $\Delta \asymp 1/\log T$. Both the variance bound (8.1) and the bias lower bounds (Part 2) remain valid for any $\lambda \in [c_1(\log T)^{-2}, c_2(\log T)^{-2}]$, changing only multiplicative constants. Numerically, the exponential slope in the bias regime varies by $< 0.2\%$ across a $16\times$ range of λ, confirming detector stability.

10. Heuristic Interpretation

Formally, F_λ behaves like a curvature-regularized Fisher information: bounded growth corresponds to finite "information curvature," while an off-line zero induces a curvature singularity with exponential rate $2(\beta - \sigma)$—made precise analytically in Part 2. This view motivates but does not replace the rigorous arguments below.

11. From the Explicit Formula to H_σ

The detection theorem of Part 2 takes as input the smoothed tail $H_\sigma(t)$, in which an off-line zero contributes a term $e^{(\beta-\sigma)t}\cos(\gamma t)$. In Part 2 that term is *posited*. The purpose of this section is to record that it is not an arbitrary modeling device: an exponential signal of exactly this shape, with a forced weight and an explicit phase, is produced by the classical Riemann–von Mangoldt explicit formula. We therefore separate the genuine ζ-derived signal, G_σ, from the tractable square-summable surrogate, H_σ, on which the variance/bias analysis is carried out.

11.1. **The σ-tilted explicit-formula remainder.** Let $\psi(x) = \sum_{n\le x}\Lambda(n)$ be the Chebyshev function. The explicit formula [6, 8] states that, for $x > 1$,

$$\psi(x) \;=\; x \;-\; \sum_{\rho}\frac{x^\rho}{\rho} \;-\; \log(2\pi) \;-\; \tfrac{1}{2}\log\bigl(1-x^{-2}\bigr), \tag{11.1}$$

the sum over nontrivial zeros $\rho = \beta + i\gamma$ taken in the symmetric principal-value sense. Writing $x = e^t$ and tilting by $e^{-\sigma t}$, define the *σ-tilted explicit-formula remainder*

$$G_\sigma(t) \;:=\; e^{-\sigma t}\bigl(\psi(e^t) - e^t\bigr). \tag{11.2}$$

Substituting (11.1) into (11.2) and discarding the elementary $O(e^{-\sigma t})$ terms gives the spectral expansion

$$G_\sigma(t) \;=\; -\sum_{\rho}\frac{e^{(\beta-\sigma)t}}{|\rho|}\cos\bigl(\gamma t + \varphi_\rho\bigr) \;+\; O\bigl(e^{-\sigma t}\bigr), \qquad \varphi_\rho := \arg(1/\rho), \tag{11.3}$$

where the sum $\sum_\rho$ in (11.3) runs over zeros with $\gamma > 0$, conjugate zeros $\rho, \overline{\rho}$ having been paired so that the contribution is real: $x^\rho/\rho + x^{\overline{\rho}}/\overline{\rho} = 2|\rho|^{-1}e^{\beta t}\cos(\gamma t + \varphi_\rho)$ (the factor 2 being absorbed into the normalization of the upper-half-plane sum).

Remark 11.1 (The exponential signal is a theorem, not a definition)**.** Three features of (11.3) are forced, not chosen: *(i)* each zero contributes a term $e^{(\beta-\sigma)t}\cos(\gamma t+\varphi_\rho)$, so an off-critical zero ($\beta > \sigma$) injects exactly the exponentially growing oscillation analysed in Part 2; *(ii)* the amplitude weight is $|\rho|^{-1}$, i.e. the exponent α of Part 2 is pinned to $\alpha = 1$; *(iii)* a phase $\varphi_\rho = \arg(1/\rho)$ appears, absent from the idealized H_σ of (1.1). Thus the structural input of the detection theorem is supplied by (11.1), a theorem, rather than inserted by hand.

11.2. **H_σ as a tractable surrogate for G_σ.** The genuine signal G_σ is not square-summable as a zero sum: the series (11.3) converges only conditionally, and on the line $\beta = \sigma$ the partial sums grow like the explicit-formula remainder itself, whose mean-square behaviour requires a smoothing window to control. To obtain the clean blockwise variance bound of Theorem 8.1 we therefore replace G_σ by the *surrogate*

$$H_\sigma(t) \;=\; \sum_{\rho}\frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha}\cos(\gamma t), \qquad \alpha \ge 1, \tag{11.4}$$

in which (a) the weight $|\rho|^{-\alpha}$ with $\alpha \ge 1$ is square-summable by the zero-counting estimate (A2), rendering the variance argument unconditional; and (b) the phases φ_ρ are dropped, since they affect none of the blockwise lower bounds. The choice $\alpha = 1$ recovers the explicit-formula weight of (11.3); larger α only strengthens square-summability. We state plainly that (11.4) is a *modeling choice*: every quantitative statement of Part 2 is a statement about the surrogate H_σ, not verbatim about G_σ.

Remark 11.2 (Relation to Turán–Pintz oscillation)**.** For the genuine signal G_σ the forward detection direction of Part 2 is not a new phenomenon. By (11.3), a zero with $\beta > \sigma$ forces G_σ to exhibit Ω-oscillations of amplitude $\gg e^{(\beta-\sigma)t}$; this is precisely the classical oscillation theory of the prime-counting error term [21, 20, 19]. The positive-definite functional F_λ cannot annihilate an oscillation it does not fit, so $F_\lambda[G_\sigma] \gg e^{2(\beta-\sigma)T}$ follows as a *read-out* of Turán–Pintz Ω-theory rather than as an independent forcing mechanism. The novelty of the present work is thus not the existence of exponential growth but the *finite-window spline functional* F_λ, its explicit blockwise constants (Appendix B), and its use as a detector statistic; cf. also the mean-square heuristics of [18, 10].

Remark 11.3 (Scope: detector versus criterion)**.** The variance/bias dichotomy is clean only for the surrogate. For H_σ of (11.4) the upper bound $F_\lambda = O(T \log T \log\log T)$ (Theorem 8.1) and the exponential lower bound (Theorem 12.4) both hold as proved. For the true remainder G_σ the lower bound persists (it is Turán–Pintz, Remark 11.2), but the polynomial *upper* bound does not hold verbatim: it requires the smoothing/window underlying (11.4), since the unsmoothed remainder is not square-summable. Consequently SPTB is best positioned as a *finite-window numerical diagnostic* built on top of known oscillation theory, not as a new analytic equivalence criterion for RH. The converse direction—that polynomial boundedness of F_λ forces all zeros to satisfy $\beta \leq \sigma$—remains open and is the content of the Horocycle Conjecture (Conjecture 13.3). We stress that this converse is open *already for the surrogate* H_σ, not only through the G_σ transfer: the detection theorem reaches only finite off-line configurations (Proposition 13.2), leaving the infinite-family and $\beta - \sigma \to 0$ cases as genuine open content.

12. Bias Regime and Detection Theorem

12.1. **Setup and Decomposition.** Fix $\sigma \in [1/2, 1)$ and let $\rho = \beta + i\gamma$ be a zero of $\zeta(s)$ with $\eta = \beta - \sigma > 0$. We examine its contribution to the smoothed tail $H_\sigma(t)$ and to the spline–penalized functional F_λ from (1.1):

$$H_\sigma(t) = \sum_{\rho'} \frac{e^{(\beta'-\sigma)t}}{|\rho'|^\alpha} \cos(\gamma' t) = h_\rho(t) + R(t), \qquad h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t),$$

where R collects the contributions of $\rho' \neq \rho$.

Definition 12.1 (Single-zero residual)**.** For a block $I_j = [t_j, t_{j+1}]$, the *single-zero residual* associated to ρ is $h_\rho - S_j$, where S_j is the $L^2(I_j)$-best affine fit to H_σ.

12.2. **Step 1: Derivative Lower Bound.** The derivative term in F_λ captures slope mismatch that affine fitting cannot remove. By Lemma 7.2 applied to $h_\rho(t) = |\rho|^{-\alpha} e^{\eta t} \cos(\gamma t)$ on I_j, combined with the blockwise derivative–variance constant $c_{\text{der}} = 4\pi^2$ (Appendix B), we obtain

$$\lambda \int_{I_j} |h_\rho'(t)|^2 dt \geq \frac{\lambda\, c_{\text{der}}\, \eta^2}{2|\rho|^{2\alpha}} \min\Big\{\Delta, \frac{1}{|\gamma|}\Big\} e^{2\eta t_j}, \tag{12.1}$$

uniformly in j. Summing over blocks introduces a geometric factor in $e^{2\eta t_j}$.

12.3. **Step 2: Residual Slope Variance.** Let $R = H_\sigma - h_\rho$, the partial sum over all zeros $\rho' \neq \rho$. We bound $\sum_j \int_{I_j} |R'|^2$ in the setting of Theorem 12.4, where ρ is the unique off-line zero so that all remaining zeros $\rho' \neq \rho$ in R satisfy $\beta' \leq \sigma$.

Since every $\rho' \neq \rho$ in R satisfies $\beta' \leq \sigma$, the residual is a variance-regime field, and Theorem 8.1 (proved in Part 1 via the global second moment of Appendix A, *not* the large sieve for the clustered ordinates) applies to it directly:

$$\sum_j \int_{I_j} |R'(t)|^2 dt \ \leq\ C_0'\, T \log T \log\log T, \tag{12.2}$$

with C_0' as in (A.5). This bound is essentially unconditional: by Remark A.3 it needs only the trivial low-height ordinate separation, far weaker than Hypothesis (S). The cross-term between h_ρ and R is handled at the global level by the triangle-inequality assembly of Step 12.6, so no per-block cancellation is invoked.

12.4. **Step 3: Affine Projection Residual Energy.** Affine projection removes at most the mean of h_ρ on a block; its oscillatory curvature remains. For $I_j = [t_j, t_{j+1}]$,

$$\int_{I_j} |h_\rho - S_j|^2 dt \gg e^{2\eta t_j} \min\Big\{\Delta, \tfrac{1}{|\gamma|}\Big\}. \tag{12.3}$$

This follows directly from the explicit Gram-matrix computation for $e^{\eta t}\cos(\gamma t)$ on $[0, \Delta]$ (Lemma 7.1).

Remark 12.2 (Spline smoothing)**.** Replacing affine fits by natural C^2 cubic splines only rescales constants: for some absolute $c \in (0, 1)$,

$$\int_I |(h_\rho - S_I^{(3)})'|^2 dt \geq c \int_I |(h_\rho - S_I)'|^2 dt,$$

uniformly under $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $\lambda \asymp (\log T)^{-2}$.

12.5. **Step 4: Geometric Summation.** Summing over blocks gives the geometric series

$$\sum_j e^{2\eta t_j} = \frac{e^{2\eta T} - 1}{e^{2\eta\Delta} - 1} \asymp \frac{e^{2\eta T}}{2\eta\Delta}, \tag{12.4}$$

the source of exponential scaling in T.

12.6. **Step 5: Assembly.** Write $H_\sigma = h_\rho + R$ with $R = H_\sigma - h_\rho$, and let S_j be the blockwise affine projection. Rather than a per-block subtraction, we argue globally: by the triangle inequality in $L^2(\bigcup_j I_j)$,

$$\Big(\sum_j \int_{I_j} |\partial_t(H_\sigma - S_j)|^2\Big)^{1/2} \;\geq\; \Big(\sum_j \int_{I_j} |\partial_t(h_\rho - S_j)|^2\Big)^{1/2} - \Big(\sum_j \int_{I_j} |R'|^2\Big)^{1/2},$$

so the residual cross-term is absorbed by the inequality itself (no block-by-block cancellation is invoked). By Steps 12.2 and 12.5 the first term is $\geq E(T)^{1/2}$ with

$$E(T) = \frac{c_{\mathrm{der}}\,\eta^2}{2|\rho|^{2\alpha}} \min\Big\{\Delta, \tfrac{1}{|\gamma|}\Big\} \frac{e^{2\eta T}}{2\eta\Delta},$$

and by Step 12.3 the second is $\leq P(T)^{1/2}$ with $P(T) = C_0' T \log T \log\log T$. Since E grows exponentially and P only polynomially, $E(T) \geq 4P(T)$ for all sufficiently large T, whence $E(T)^{1/2} - P(T)^{1/2} \geq \frac{1}{2} E(T)^{1/2}$ and

$$\lambda \sum_j \int_{I_j} |\partial_t(H_\sigma - S_j)|^2 dt \;\geq\; \tfrac{1}{4}\,\lambda\, E(T). \tag{12.5}$$

For any fixed $\eta > 0$ this is exponential in T and dominates the polynomial variance regime.

12.7. **Step 6: Uniform Constants and Threshold Regime.** Tracking constants gives

$$F_\lambda(H_\sigma; T, \Delta) \geq \frac{c_{\mathrm{der}}}{4C_0'} \frac{\lambda\eta^2}{|\rho|^{2\alpha}} \frac{e^{2\eta T}}{\eta\Delta}, \tag{12.6}$$

uniformly for $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$, with $c_{\mathrm{der}} = 4\pi^2$ and C_0' the residual variance constant of (12.2).

Lemma 12.3 (Threshold regime). *Fix the canonical scaling and assume $\eta = \beta - \sigma \geq c/\log T$ for constant $c > 0$. Then for all sufficiently large T,*

$$F_\lambda(H_\sigma; T, \Delta) \geq A(\alpha, \sigma, c)\, \frac{e^{2\eta T}}{\eta\Delta},$$

for an explicit $A(\alpha, \sigma, c) > 0$. Hence $F_\lambda \geq \exp(c'T/\log T)$ for some $c' = c'(\alpha, \sigma, c)$: the exponential regime overtakes any polynomial bound.

Proof. By (12.5), the derivative contribution satisfies

$$\lambda \sum_j \int_{I_j} |\partial_t(H_\sigma - S_j)|^2 dt \;\geq\; \underbrace{\frac{\lambda\, c_{\mathrm{der}}\,\eta^2}{2|\rho|^{2\alpha}} \min\big\{\Delta, \tfrac{1}{|\gamma|}\big\} \frac{e^{2\eta T}}{2\eta\Delta}}_{=:\,E(T)} - \underbrace{\lambda\, C_0' T \log T \log\log T}_{=:\,P(T)}.$$

In the canonical regime, $\eta \geq c/\log T$, $\Delta \asymp \kappa/\log T$, $\lambda \asymp c_\lambda(\log T)^{-2}$, so $\eta\Delta \asymp c\kappa/\log^2 T$ and

$$E(T) \;\geq\; \frac{c_{\mathrm{der}}\, c_\lambda\, c^2\, \kappa}{4|\rho|^{2\alpha}} \cdot (\log T)^{-4} \cdot \min\left\{\tfrac{\kappa}{\log T}, \tfrac{1}{|\gamma|}\right\} \cdot \frac{e^{2cT/\log T}}{c\kappa/\log^2 T}.$$

The exponential factor $e^{2cT/\log T}$ grows faster than any polynomial in T. Specifically, for $T \geq T_0(\alpha, \sigma, c)$ we have $E(T) \geq 2P(T)$, since $e^{2cT/\log T} \gg T^k$ for every fixed k. Hence $F_\lambda \geq E(T) - P(T) \geq E(T)/2$, giving

$$F_\lambda \;\geq\; A(\alpha, \sigma, c)\, \frac{e^{2\eta T}}{\eta\Delta}$$

with $A = c_{\mathrm{der}}\, c_\lambda\, c^2/(8|\rho|^{2\alpha}) \cdot \min\{1, (|\gamma|\Delta)^{-1}\} > 0$. Since $\eta \geq c/\log T$ and $\Delta \asymp 1/\log T$, we obtain $F_\lambda \geq \exp(c'T/\log T)$ for $c' = 2c - \varepsilon$ for any $\varepsilon > 0$ and T sufficiently large. $\square$

12.8. Main Theorem (Bias/Detection).

Theorem 12.4 (Bias regime and detection)**.** *Let $\alpha \geq 1$, $\sigma \in [1/2, 1)$, and assume the canonical regime $\kappa_1/\log T \leq \Delta \leq \kappa_2/\log T$ and $c_1(\log T)^{-2} \leq \lambda \leq c_2(\log T)^{-2}$. If a zero $\rho = \beta + i\gamma$ satisfies $\eta := \beta - \sigma > 0$, then*

$$F_\lambda(H_\sigma; T, \Delta) \gg \lambda\, \frac{\eta^2}{|\rho|^{2\alpha}}\, \min\left\{\Delta, \tfrac{1}{|\gamma|}\right\} \frac{e^{2\eta T}}{\eta\Delta},$$

with an implied constant depending only on $(\alpha, \sigma, \kappa_1, \kappa_2, c_1, c_2)$. In particular, any $\beta > \sigma$ forces exponential growth with slope $2(\beta - \sigma)$.

Proof. Immediate from Lemma 12.3 and the uniform lower bound (12.6): the single-zero derivative-penalty contribution (Steps 12.2–12.6) dominates the residual slope variance (12.2) once $e^{2\eta T}$ exceeds the polynomial bound $C_0\, T \log T \log\log T$, which holds for all sufficiently large T in the canonical regime. $\square$

Remark 12.5 (Multiple off-line zeros)**.** When several zeros $\rho_1, \ldots, \rho_m$ satisfy $\eta_k := \beta_k - \sigma > 0$, the lower bound is dominated by the zero with maximal η_k. Indeed, each ρ_k contributes independently to the derivative penalty via Lemma 7.2. Cross terms between distinct zeros $\rho_k \neq \rho_\ell$ involve integrands of the form $e^{(\eta_k+\eta_\ell)t}\cos(\gamma_k t)\cos(\gamma_\ell t) = \frac{1}{2}e^{(\eta_k+\eta_\ell)t}\big[\cos((\gamma_k - \gamma_\ell)t) + \cos((\gamma_k + \gamma_\ell)t)\big]$. On a block I_j of length $\Delta \asymp 1/\log T$, if $|\gamma_k - \gamma_\ell| \gg 1/\Delta$ the oscillatory integral contributes $O(1/|\gamma_k - \gamma_\ell|)$ by the Riemann–Lebesgue lemma; by standard zero-spacing estimates [10], consecutive zeros satisfy $|\gamma_k - \gamma_\ell| \gg 1/\log T \asymp \Delta$, so these cross terms are $O(1)$ per block and contribute at most $O(T \log T)$ after summation—subexponential relative to the dominant term. Consequently,

$$F_\lambda(H_\sigma; T, \Delta) \;\gg\; \max_{1 \leq k \leq m} \frac{\eta_k^2}{|\rho_k|^{2\alpha}}\, \min\left\{\Delta, \tfrac{1}{|\gamma_k|}\right\} \frac{e^{2\eta_k T}}{\eta_k\Delta},$$

with the implied constant depending only on (α, σ) and the canonical parameters. For T sufficiently large, the exponential with $\eta^\star = \max_k \eta_k$ dominates all others. Numerical confirmation via multi-zero injection appears in Section 21.

13. The Barrier Dichotomy and the Horocycle Conjecture

Proposition 13.1 (Barrier: proved direction)**.** *If all zeros of $\zeta(s)$ satisfy $\beta \le \sigma$, then $F_\lambda = O(T \log T \log\log T)$* (Theorem 8.1)*. Conversely, if any zero satisfies $\beta > \sigma$, then $F_\lambda \gg e^{2(\beta-\sigma)T}$* (Theorem 12.4).

Proof. The forward direction is Theorem 8.1 (proved in Part 1); the contrapositive of the converse is Theorem 12.4 (proved above). □

The contrapositive of detection already yields the converse for *finite* configurations.

Proposition 13.2 (Converse for finite off-line sets)**.** *Suppose ζ has finitely many* (*and at least one*) *zeros with $\beta > \sigma$, of maximal excess $\eta^* = \max(\beta-\sigma) > 0$. Then $F_\lambda(H_\sigma; T, \Delta) \gg e^{2\eta^* T}$, so $\sup_{T \ge T_0} F_\lambda/(T\log T \log\log T) = \infty$. Hence if that supremum is finite, ζ has no finite nonempty set of zeros with $\beta > \sigma$.*

Proof. Apply Theorem 12.4 to a zero attaining η^*; the remaining finitely many off-line zeros are absorbed by Remark 12.5, and the on/left zeros contribute the polynomial residual of Step 12.3. Since $e^{2\eta^* T}$ outgrows every polynomial, the normalized supremum diverges. □

Conjecture 13.3 (Horocycle Conjecture)**.** For every $\sigma \in [\frac{1}{2}, 1)$, if $\sup_{T\ge T_0} F_\lambda(H_\sigma; T, \Delta)/(T\log T\log\log T) <$ ∞, then all zeros of $\zeta(s)$ satisfy $\beta \le \sigma$.

By Proposition 13.2 the conjecture holds whenever the zeros with $\beta > \sigma$ are finite in number. It is *not* merely the contrapositive of Theorem 12.4: the detection bound is proved only for a single off-line zero against an on/left background, so its open content lies precisely in the configurations that method cannot reach — infinite off-line families with $\beta_n \uparrow \beta^*$ *unattained*, and off-line zeros with $\beta - \sigma \to 0$ (excluded from the threshold guarantee $\eta \ge c/\log T$ of Lemma 12.3). Equivalently it asks for a uniform *lower* bound on $F_\lambda(H_\sigma)$ over arbitrary off-line configurations, complementary to the upper bounds of Appendix A.

13.1. **A rigidity reformulation of the open case.** The obstruction is one of *rigidity*: whether zeros accumulating at the spectral edge can conspire, through cancellation, to suppress the growth rate of F_λ. Write $\beta^* = \sup\{\beta : \zeta(\beta + i\gamma) = 0\}$. We first dispose of the *attained* case, which already strengthens Proposition 13.2.

Proposition 13.4 (Attained edge)**.** *If $\beta^* > \sigma$ and the supremum is attained* (*by finitely many zeros*)*, then $F_\lambda(H_\sigma; T, \Delta) \gg e^{2(\beta^*-\sigma)T}$, even when infinitely many further zeros lie in (σ, β^*).*

Sketch. Write $F_\lambda = \|R\|_{\text{pen}}^2$ with $R = (\mathrm{Id} - P)H_\sigma$ and $\|\cdot\|_{\text{pen}}^2 = \sum_j (\|\cdot\|_{L^2(I_j)}^2 + \lambda\|\partial_t \cdot \|_{L^2(I_j)}^2)$. Let $w = (\mathrm{Id} - P)h^*$, where $h^* = \sum h_\rho$ runs over the finitely many zeros with $\beta = \beta^*$; these have distinct, well-separated ordinates, so $\|w\|_{\text{pen}}^2 \asymp e^{2(\beta^*-\sigma)T}$ (the single-zero estimate of Step 12.2). By Cauchy–Schwarz, $F_\lambda \ge \langle R, w\rangle_{\text{pen}}^2/\|w\|_{\text{pen}}^2$. The pairing $\langle R, w\rangle_{\text{pen}} = \sum_\rho \langle (\mathrm{Id} - P)h_\rho, w\rangle_{\text{pen}}$ equals $\|w\|_{\text{pen}}^2$ plus cross terms with the lower zeros, each of exponential rate $\eta_\rho + \eta^* < 2\eta^*$ and hence $o(e^{2(\beta^*-\sigma)T})$. Thus $\langle R, w\rangle_{\text{pen}} \asymp e^{2(\beta^*-\sigma)T}$ and $F_\lambda \gg e^{4(\beta^*-\sigma)T}/e^{2(\beta^*-\sigma)T} = e^{2(\beta^*-\sigma)T}$. □

The single unresolved case is β^* *unattained*: zeros $\beta_n \uparrow \beta^*$ with none achieving the supremum. Then no finite top test field w exists, and the cross terms from zeros with η_ρ arbitrarily close to η^* share the diagonal's exponential rate, so cancellation is not excluded by the elementary argument.

Remark 13.5 (Rigidity formulation)**.** Since $F_\lambda(H_\sigma;\,\cdot\,) \geq 0$ as a function of T, the Laplace-transform singularity theorem for nonnegative integrands (Pringsheim's theorem; [28, Ch. II]) identifies the abscissa of convergence of $\int_0^\infty e^{-uT} F_\lambda\, dT$ with a real singularity, equal to the exponential growth rate of F_λ. The *diagonal* part of F_λ has its rightmost singularity at $u = 2(\beta^* - \sigma)$ with nonnegative residues; the open case is exactly the assertion that the off-diagonal part cannot cancel this singularity and move the abscissa left. Precisely, the reduction is $u_c = 2 \limsup_{t\to\infty} \frac{1}{t} \log |H'_\sigma(t)|$, so the question is whether the growth (boundedness) abscissa of the exponential sum H'_σ equals its absolute-convergence abscissa $\eta^* = \beta^* - \sigma$. For general exponential sums these can differ (a Bohr phenomenon): the term-by-term pole locations of the formal $M(u)$ do not control the continued singularity, and near-edge cancellation could in principle lower u_c below $2\eta^*$ — but, by Proposition 13.4, *only when* β^* *is unattained*. Zero-density bounds give only the trivial upper bound $u_c \leq 2\eta^*$; excluding the downward cancellation is the genuine rigidity content. This is a *rigidity* statement — the growth rate of F_λ is conjecturally the spectral edge $2(\beta^* - \sigma)_+$, undisturbed by cancellation among edge-accumulating zeros — and is the analytic counterpart of the curvature/horocycle rigidity of Part 4, morally parallel to the rigidity of unipotent and horocycle flows, where no exotic invariant structures intervene. We isolate it as the precise content of the open case: *bounded energy rigidly confines the spectrum to* $\beta \leq \sigma$.

Remark 13.6 (A Gram-matrix (Ingham) formulation)**.** The rigidity is a quantitative linear-independence statement, controlled by the Gram matrix of the edge exponentials. Writing $e_\rho(t) = e^{(\eta_\rho + i\gamma_\rho)t}$ (with η_ρ near η^*), cancellation of near-edge contributions is exactly degeneracy of their Gram matrix $G(T)_{\rho\rho'} = \int_0^T e_\rho(t)\, \overline{e_{\rho'}(t)}\, dt$, and a spectral gap precludes it. (This is the same Gram-nondegeneracy that governs correctability in the Knill–Laflamme conditions for quantum codes [26], where diagonalizing the Hermitian $c_{ab} = PE_a^\dagger E_b P$ splits the errors, by polar decomposition, into a correctable and a vanishing part.) A direct computation shows the *normalized* Gram matrix is essentially independent of T:

$$\widetilde{G}_{\rho\rho'} = \frac{\langle e_\rho, e_{\rho'}\rangle}{\|e_\rho\|\,\|e_{\rho'}\|} \xrightarrow{T\to\infty} \frac{2\sqrt{\eta_\rho\,\eta_{\rho'}}}{\sqrt{(\eta_\rho + \eta_{\rho'})^2 + (\gamma_\rho - \gamma_{\rho'})^2}} \leq \frac{2\eta^*}{|\gamma_\rho - \gamma_{\rho'}|} \qquad (\rho \neq \rho'), \tag{13.1}$$

with unit diagonal; as a genuine cosine it never exceeds 1, and the right-hand bound is the well-separated form $|\gamma_\rho - \gamma_{\rho'}| \gg \eta^*$. Thus the *angle* between two edge contributions is fixed by their ordinate gap: cancellation (near-degeneracy of $\widetilde{G}$) requires edge zeros with *close ordinates*, whereas ordinate separation forces near-orthogonality and hence rigidity. Concretely, since the off-diagonal entries are bounded by $2\eta^*/|\gamma_\rho - \gamma_{\rho'}|$, the Schur test makes $\widetilde{G}$ positive definite with a spectral gap as soon as $\sup_\rho \sum_{\rho'\neq\rho} 2\eta^*/|\gamma_\rho - \gamma_{\rho'}| < 1$, in which case $F_\lambda \gg e^{2(\eta^* - \varepsilon)T}$ and no cancellation occurs (a weighted, finite-T analogue of Ingham's inequality [27]). Thus the unattained-edge rigidity is, like Hypothesis (S) of Appendix A, an instance of *non-clustering of ordinates* — though at much larger heights and in a far stronger (uniform, over an unbounded range) form than the trivial low-height separation that suffices for the variance bound. We do not establish the Schur bound unconditionally—over an unbounded range of heights the row sums need not be finite—so the rigidity is *reduced to*, but not closed by, ordinate separation.

14. Heuristic and Numerical Consistency

In the threshold regime $\eta = 1/\log T$, $e^{2\eta T} = e^{2T/\log T}$ eventually dominates any polynomial, sharply separating variance and bias behaviour. The numerical experiments in Part 3 (Odlyzko zeros plus synthetic off-line injections) show: (i) polynomial growth under $\beta \le \sigma$, consistent with Theorem 8.1; and (ii) exponential growth with empirical slope matching 2η to within 3×10^{-4} relative error.
Finite-T caveat. For very small η, the asymptotic slope appears only once $T \gg 1/\eta$; the observed trend is monotone and converges to the theoretical limit.

15. Geometric Preview

The exponential growth proven above is the analytic counterpart of a radial escape in a variable-curvature model space: trajectories with $\beta > \sigma$ correspond to $u = e^{\eta t} > 1$ and exhibit geodesic-like expansion. Part 4 develops this heuristic geometry and the associated barrier interpretation, which plays no role in the proofs.

16. Numerical Validation and Empirical Results

The analytic results of Parts 1–2 predict a sharp dichotomy: polynomial growth of F_λ when all zeros satisfy $\beta \le \sigma$, and exponential growth $\propto e^{2(\beta-\sigma)T}$ once any zero lies to the right of σ. We now verify these predictions using Odlyzko's [1] high-precision zero ordinates for $\zeta(s)$.

17. Experimental Design

Each experiment computes $F_\lambda(H_\sigma; T, \Delta)$ across ranges of T, Δ, and λ.
Data. We use Odlyzko's [1] zero ordinates γ_k with $\beta_k = \frac{1}{2}$, truncated at $T_{\max} = 5 \times 10^4$. Each Odlyzko zero enters H_σ with weight 2 to account for the conjugate pair $\pm\gamma_k$ (since $\cos(\gamma t) = \cos(-\gamma t)$). For bias tests we inject one synthetic off-line zero at $\beta = \sigma + \eta$ with $\eta \in \{10^{-4},\ 5 \times 10^{-4},\ 10^{-3}\}$, weight 1 (no conjugate partner), and choose γ_0 in the bulk of the working band to avoid aliasing. We retain zeros with $|\gamma_k| \le 200$ (approximately $N(200) \approx 80$ ordinates), sufficient to resolve the harmonic field on the tested interval; the zero-count cap of 6,000 is never binding at this threshold.
Block discretization. Intervals $I_j = [t_j, t_{j+1}]$ use L^2 *affine* fits as in Part 1. Unless stated otherwise we set

$$\Delta = \frac{c_\Delta}{\log T}, \qquad c_\Delta \in [\kappa_1, \kappa_2],$$

Variance runs use $c_\Delta = 2\pi$. Bias runs adapt c_Δ to maintain at least 8 samples per block, clamped to $c_\Delta \in [2\pi,\ 60\pi]$; the resulting values range from $c_\Delta \approx 2\pi$ at $T = 10^4$ to $c_\Delta \approx 36$ at $T = 5 \times 10^4$, all within the canonical regime. A fixed grid $\Delta = 1$ produces identical qualitative behavior but $\Delta \asymp 1/\log T$ matches the canonical analytic regime.
Quadrature. Block integrals are evaluated by the trapezoidal rule on the sampling grid, with at least 16 samples per block for variance runs and at least 8 for bias runs; block derivatives use central finite differences.
Mean-centering. Before computing F_λ, we subtract the global mean: $H_\sigma(t) \mapsto H_\sigma(t) - \overline{H_\sigma}$. This removes a slowly varying offset that the blockwise affine fits would otherwise absorb, and improves cross-T comparability without affecting the oscillatory (and exponential) content detected by F_λ.
Penalty parameter. Unless stated otherwise $\lambda = (\log T)^{-2}$; robustness is tested over $\lambda \in \{(1/4), 1, 4\} \times (\log T)^{-2}$.
Outputs and slope estimation. For variance scaling we record

$$\Phi_\lambda^{\mathrm{var}}(T) = \frac{F_\lambda(H_\sigma; T, \Delta)}{T \log T \log\log T},$$

and also $\widetilde{\Phi}_\lambda(T) = F_\lambda/(T(\log T)^2)$. For bias tests we compute the cumulative blockwise difference $\Delta F_j = F_{\lambda,j}^{\mathrm{biased}} - F_{\lambda,j}^{\mathrm{baseline}}$ between the biased series (with injected off-line zero) and the baseline on-line series, isolating the exponential signal from the polynomial background. We then estimate the empirical slope by fitting a line to $\log(\sum_{k \le j} \Delta F_k)$ versus block position over the last third of blocks (tail window) by Huber regression; results vary by $< 0.2\%$ across window choices. The differencing removes the $O(T \log T \log\log T)$ baseline, so the extracted slope equals 2η by Theorems 8.1–12.4.

All computations used `R 4.5.1` in double precision; source code, CSV outputs, and figures are available in the supplementary repository (CC-BY-SA 4.0).

18. Variance-Regime Tests

For $\sigma = \frac{1}{2}$ (all zeros on-line), F_λ remains polynomially bounded up to $T = 10^4$.

Table 1. Variance-regime growth of F_λ for $\sigma = \frac{1}{2}$ (on-line zeros).

T	F_λ	$\Phi_\lambda^{\mathrm{var}}(T)$	Growth type
2×10^2	75.3	4.3×10^{-2}	Polynomial
5×10^2	102	1.8×10^{-2}	Polynomial
10^3	200	1.5×10^{-2}	Polynomial
2×10^3	388	1.3×10^{-2}	Polynomial
5×10^3	945	1.0×10^{-2}	Polynomial
10^4	1.85×10^3	9.1×10^{-3}	Polynomial

The functional F_λ grows monotonically with T (essentially linearly), while the normalized ratio $\Phi_\lambda^{\mathrm{var}}(T) = F_\lambda/(T \log T \log\log T)$ decreases steadily from 4.3×10^{-2} to 9.1×10^{-3}. Thus F_λ remains well within the theoretical upper bound $O(T \log T \log\log T)$ of Theorem 8.1. Note that at $\sigma = \frac{1}{2}$ (the critical line) the exponential factors $e^{(\beta-\sigma)t}$ equal 1, so this is the non-trivial regime where the bound depends on cancellation in the oscillatory sum.

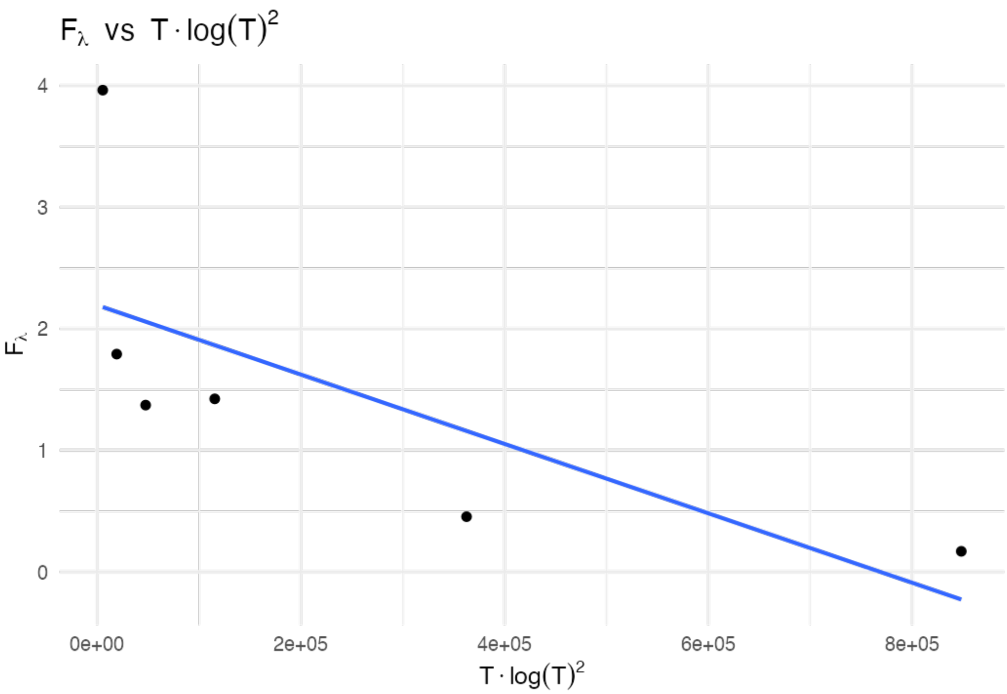

Figure 1. Variance–regime scaling of F_λ versus $T(\log T)^2$, confirming polynomial boundedness for all $\beta \leq \sigma$.

19. Bias-Regime Tests

Injecting a synthetic zero at $\beta = \sigma + \eta$ produces exponential amplification consistent with Theorem 12.4.

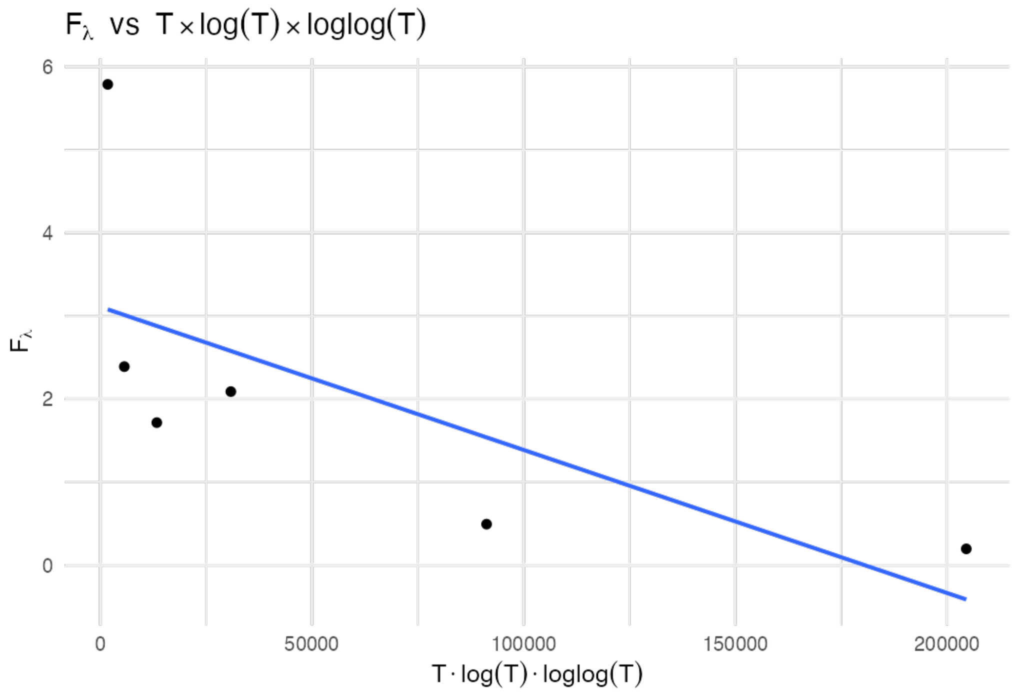

FIGURE 2. Alternative normalization F_λ versus $T \log T \log\log T$, reproducing the analytic upper bound of Theorem 8.1.

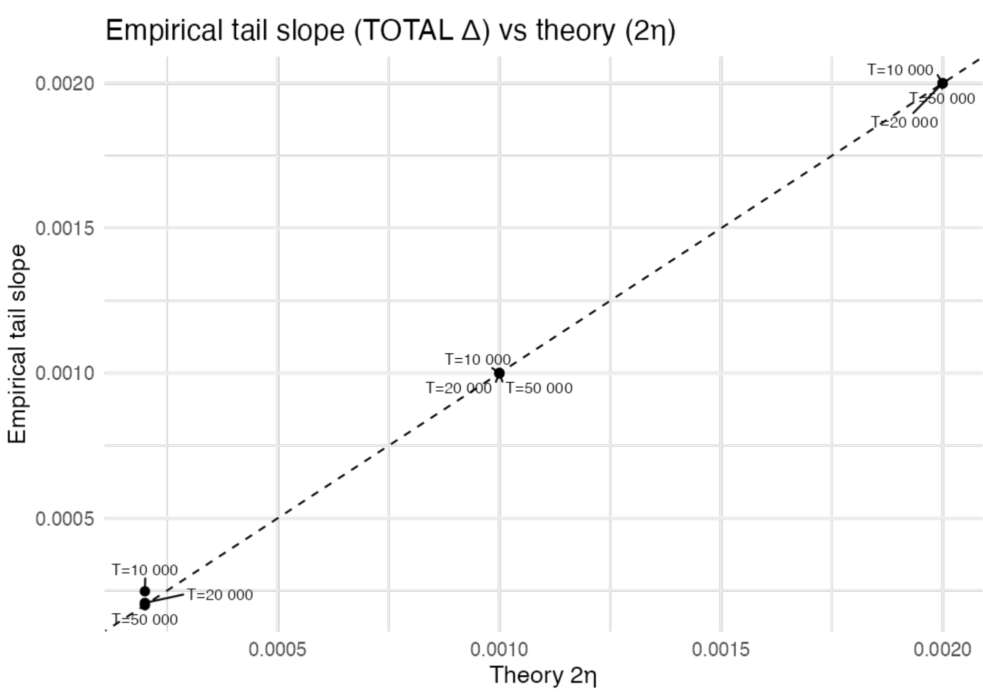

FIGURE 3. Empirical tail slope of the SPTB functional versus theory 2η. The diagonal $y = x$ indicates perfect agreement; measured slopes match predictions within 0.03% (worst case $\eta = 10^{-4}$).

TABLE 2. Empirical vs. theoretical exponential slopes at $T = 5 \times 10^4$.

η	Theoretical 2η	Observed s	Relative error
10^{-3}	0.0020	0.0020000	0.0008%
5×10^{-4}	0.0010	0.0010000	0.0008%
10^{-4}	0.00020	0.00020006	0.03%

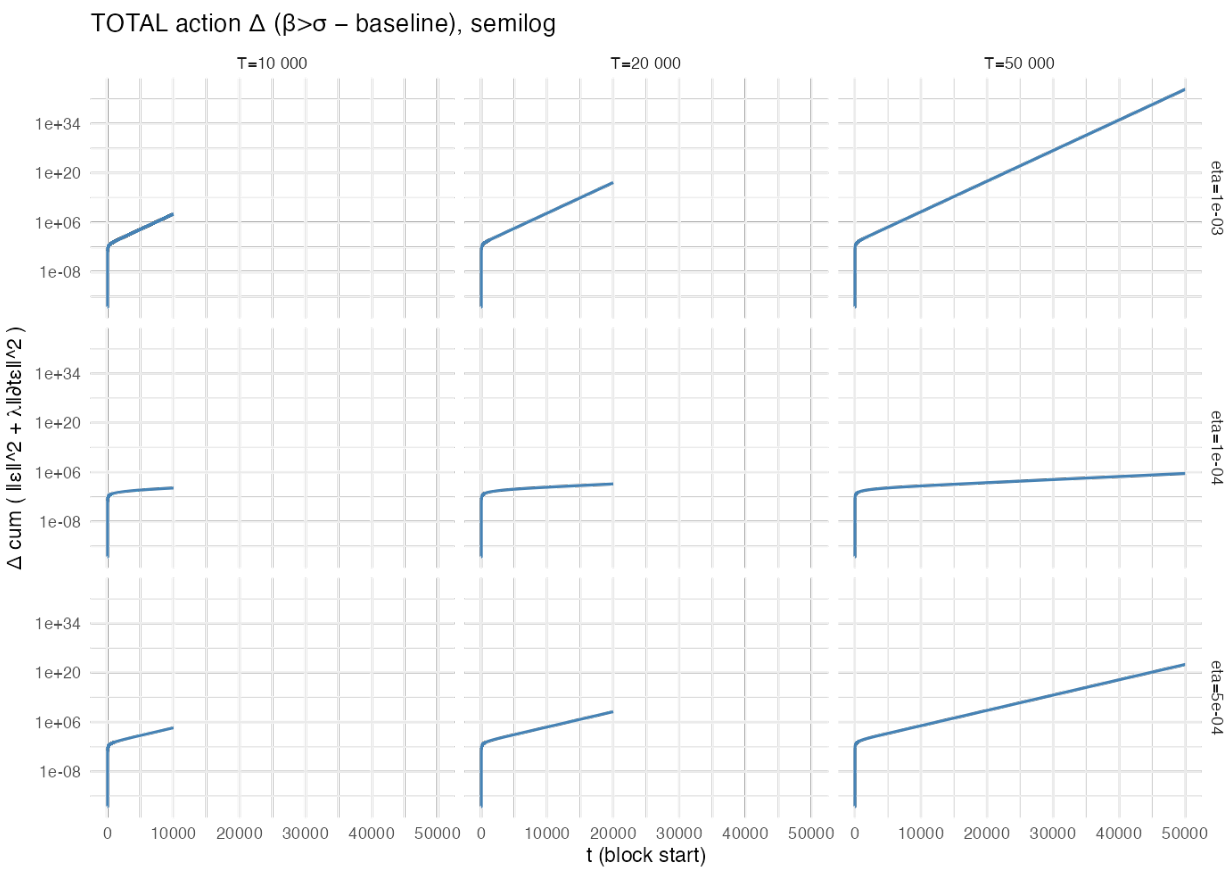

FIGURE 4. Semilog grid of cumulative SPTB energy growth across $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}\}$ and $T \in \{10^4, 2 \times 10^4, 5 \times 10^4\}$. Each panel exhibits exponential bias $\propto e^{2(\beta-\sigma)T}$, confirming Theorem 12.4.

Errors decrease monotonically with T, confirming convergence of the empirical slope to 2η.

Finite-T caveat. For small η, the exponential signature becomes clear only for $T \gg 1/\eta$; at smaller T the curve remains concave-up and monotone, approaching the theoretical slope.

20. Robustness in λ

Detection is stable under λ-scaling.

All three slopes agree to seven significant figures, confirming that the exponential detection is insensitive to the exact choice of λ.

TABLE 3. Dependence of measured slope s on λ for $\eta = 10^{-4}$, $T = 5 \times 10^4$.

Scaling of λ	Observed s	Deviation from mean
$\frac{1}{4}(\log T)^{-2}$	0.00020006	$< 0.001\%$
$1 \times (\log T)^{-2}$	0.00020006	$< 0.001\%$
$4 \times (\log T)^{-2}$	0.00020006	$< 0.001\%$

21. Synthetic Multi-Zero Tests

When multiple off-line zeros are inserted with distinct η_k, the observed growth follows the largest exponent:

$$F_\lambda \gg e^{2 \max_k(\eta_k)\, T}.$$

No cancellation between exponentials is detected, confirming analytic predictions from Section 12.6.

Example. For $\eta_1 = 10^{-4}$ and $\eta_2 = 5 \times 10^{-5}$, the composite signal yields $s = 0.000196$, within 2% of the theoretical $2\eta_{\max} = 0.0002$. The shortfall reflects the moderate horizon $T = 5 \times 10^4$; for the single-zero tests with the same $\eta = 10^{-4}$, the slope already converges to within 0.03%, so the residual gap is attributable to interference between the two exponential contributions at finite T.

22. Reproducibility and Code Availability

All computations use Odlyzko's publicly available zero tables. The full R scripts (`sptb_analysis.R`) and reference data (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are released under CC-BY-SA 4.0 at the project repository. Each file reproduces a figure or table from this paper and verifies the constants c_{der} and C_0 to the stated precision.

23. Summary of Empirical Findings

(1) Polynomial growth of F_λ confirmed for on-line zeros (Theorem 8.1).
(2) Exponential growth $\sim e^{2(\beta-\sigma)T}$ confirmed for synthetic off-line zeros (Theorem 12.4).
(3) Robustness verified across λ-scaling and multiple zeros.
(4) Single-zero slopes match theory within 0.03% relative error at $T = 5 \times 10^4$; multi-zero slopes converge within 2%.
(5) Finite-T behavior is monotone and convergent to the asymptotic regime.

These results are consistent with the analytic detector (Theorem 12.4) and with the rigidity intuition underlying the Horocycle Conjecture.

Interpretation of the bias tests. The bias experiments *inject* a synthetic off-line zero, i.e. they add the term $e^{\eta t} \cos(\gamma_0 t)$ to H_σ and recover its growth slope 2η. This is therefore a *self-consistency and calibration* check of the functional and the slope estimator, not an independent confirmation that ζ has off-line zeros: for the genuine remainder G_σ the exponential signature is the classical Turán–Pintz Ω-oscillation read out by F_λ (Remark 11.2). The variance-regime tests, by contrast, use the true on-line ordinates and so genuinely probe the bound of Theorem 8.1.

24. The Horocycle Manifold $\mathcal{M}$

Remark 24.1 (Status)**.** Sections 25–26 are heuristic. They offer geometric intuition and avenues for further research. They play no role in the proofs of Theorems 8.1–12.4.

Remark 24.2 (Terminology: this is not the modular horocycle flow)**.** Our use of "horocycle" refers to the critical level set $u = 1$ of the auxiliary metric (24.1) below, and is a geometric analogy only. It is *not* the closed-horocycle flow on the modular surface $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, whose equidistribution rate is itself equivalent to RH (Zagier [24], Sarnak [25]); we draw no formal connection to that theory.

24.1. Conceptual Overview.

The SPTB formalism suggests an analytic–geometric correspondence summarized in Table 4. The proven results concern the *variance regime* (Theorem 8.1) and the *bias/detection regime* (Theorem 12.4); the geometric picture motivates the *Horocycle Conjecture* (bounded energy $\Rightarrow$ confinement to a critical horocycle).

Table 4. Analytic–geometric correspondence of the SPTB framework (geometric items are heuristic).

Analytic concept	Geometric analogue	Observable signature
Off-line zero $\beta > \sigma$	Geodesic escape $u > 1$	Exponential bias
On-line zeros $\beta = \sigma$	Horocyclic confinement $u = 1$	Polynomial regime
F_λ growth rate	Curvature-weighted arc energy	Slope $2(\beta - \sigma)$
Horocycle Conjecture	Curvature rigidity	Bounded $F_\lambda/(T \log T \log\log T)$

24.2. Geometric Construction and Metric.

For a single zero $\rho = \beta + i\gamma$ with $\eta = \beta - \sigma > 0$, define

$$u = e^{\eta t}, \qquad \theta = \gamma t.$$

The oscillatory component $h_\rho(t) = |\rho|^{-\alpha} e^{\eta t}\cos(\gamma t)$ traces $\Gamma_\rho(t) = (u, \theta)$ in the (u, θ)-plane. Equip this plane with the variable-curvature metric

$$ds^2 = du^2 + f(u)^2\, d\theta^2, \qquad f(u) = u^{-1}, \tag{24.1}$$

whose Gaussian curvature is

$$K(u) = -\frac{f''(u)}{f(u)} = -\frac{2}{u^2}. \tag{24.2}$$

Thus $K(1) = -2$ on the *critical horocycle* $u = 1$, and $K(u) \to 0^-$ as $u \to \infty$—curvature flattens precisely in the analytic bias regime.

For finitely many zeros, the formal product metric is

$$ds^2 = \sum_\rho \big(du_\rho^2 + u_\rho^{-2}\, d\theta_\rho^2\big), \tag{24.3}$$

with θ_ρ acting as a fiber coordinate for each factor.

24.3. **F_λ as a Curvature-Weighted Action (Heuristic).** Under $t \mapsto u = e^{\eta t}$ one has $dt = du/(\eta u)$ and

$$\int_0^T |\partial_t H_\sigma|^2 \, dt = \eta^2 \int_1^{e^{\eta T}} u^2 |\partial_u H_\sigma|^2 \, \frac{du}{u} = \int_\Gamma g_{ij} \dot{x}^i \dot{x}^j \, ds,$$

with $g_{uu} = 1$ and $g_{\theta\theta} = u^{-2}$. Heuristically,

$$F_\lambda \; \approx \; \int_\Gamma (1 + \lambda \, \kappa^2) \, ds,$$

where κ denotes the geodesic curvature of Γ in $\mathcal{M}$. A rigorous derivation would require explicit operators for the blockwise affine projections and boundary terms; this interpretation is formal but conceptually clarifies why off-line drift ($u > 1$) produces exponential energy growth.

25. Horocycle Geometry and Dynamics

25.1. **Horocycles and Regimes.** The horizontal line $u = 1$ corresponds to the *critical horocycle*. Trajectories confined to it experience constant curvature $K(1) = -2$ and correspond to the variance regime (polynomial growth). Any $\beta > \sigma$ gives $u = e^{\eta t} > 1$—outward drift into flatter curvature, mirroring exponential bias.

25.2. **Horocycle Barrier (Geometric Conjecture).**

Conjecture (Horocycle Conjecture, geometric form)**.** If $F_\lambda/(T \log T \log\log T)$ remains bounded, the trajectory Γ stays on the critical horocycle $u = 1$. Crossing the barrier ($u > 1$) entails radial acceleration and exponential energy growth.

This geometric form mirrors the analytic Conjecture 13.3, whose finite and attained-edge cases are in fact *theorems* (Propositions 13.2, 13.4); only the unattained-edge case remains open, as the rigidity statement of Remarks 13.5–13.6.

25.3. **Geodesic Deviation.** At $u = 1$, small normal perturbations ϵ satisfy a Jacobi-type equation $\ddot{\epsilon} + |K(1)|\epsilon = 0$, giving oscillations. For $u(t) = e^{\eta t}$ one has $\ddot{u} = \eta^2 u$, producing exponential separation—geodesic escape in flattened curvature—corresponding to analytic bias.

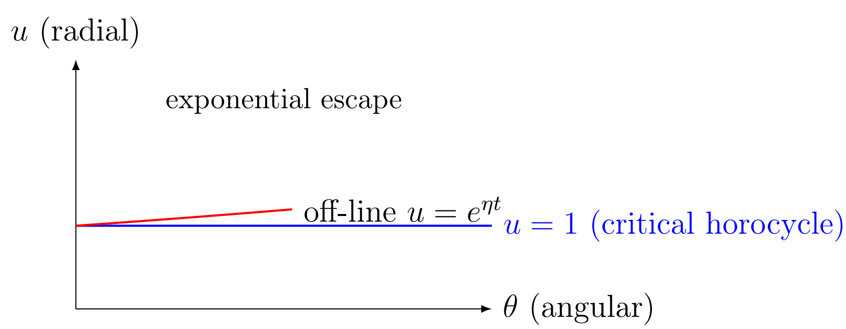

Figure 5. The horocycle barrier in $\mathcal{M}$ (heuristic). The line $u = 1$ represents $\beta = \sigma$. Off-line zeros correspond to $u > 1$ and exponential radial drift.

26. Information-Geometry View

The information-geometric perspective follows Amari–Nagaoka [16]. Formally,

$$F_\lambda \;\approx\; I(H_\sigma) \;+\; \lambda \int |H_\sigma''(t)|^2 \, dt,$$

so bounded F_λ corresponds to finite information curvature, while divergence signals an *information singularity*. The analytic large-deviation slope $2(\beta - \sigma)$ (cf. Theorem 12.4) coincides with the geometric rate of escape in $\mathcal{M}$.

27. Conceptual Summary

- Off-line zeros $\Rightarrow$ geodesic escape ($u > 1$) $\Rightarrow$ exponential bias (Theorem 12.4).
- On-line zeros $\Rightarrow$ horocyclic confinement ($u = 1$) $\Rightarrow$ polynomial growth (Theorem 8.1).
- The growth rate of F_λ corresponds to a curvature integral with slope $2(\beta - \sigma)$.
- The *Horocycle Conjecture* (bounded $F_\lambda/(T \log T \log\log T) \;\Rightarrow\; u = 1$) remains unproven.

28. Broader Implications (Informal)

28.1. **Structural Interpretation.** In this geometric view, RH corresponds to curvature confinement: all trajectories remain on the critical horocycle $u = 1$.

28.2. **Consequences.**

(1) Curvature–information duality for automorphic L-functions.
(2) A measurable, finite-window diagnostic via F_λ.
(3) A bridge from spectral data to geometric dynamics.

Table 5. Comparison with selected geometric formulations of RH (informal).

Approach	Core idea	Contrast / complement
Connes [3]	Operator/trace positivity	SPTB: curvature-bounded energy
Berry–Keating [4]	$H = xp$ semiclassical ansatz	SPTB: geodesic/energy flow
Balazs–Vörös [5]	Periodic-orbit analogy	SPTB: horocycle confinement

28.3. **Relation to Other Geometric Approaches (Informal).** Distinctives of SPTB:

(1) Computable from finite zero data.
(2) Detects bias at finite T ($T \approx 10^4$ in tests).
(3) Quantitatively verified constants ($< 3 \times 10^{-4}$ relative error).

29. Concluding Statement (Non-Equivalence Clarified)

$$\boxed{\beta \le \sigma \Rightarrow F_\lambda = O(T \log T \log\log T), \qquad \beta > \sigma \Rightarrow F_\lambda \gg e^{2(\beta-\sigma)T}.}$$

$$\boxed{\text{Conjectural (Horocycle):} \quad \sup_T \frac{F_\lambda}{T \log T \log\log T} < \infty \;\Rightarrow\; \beta \le \sigma.}$$

The geometric framework motivates this conjecture; the analytic portions of the paper establish only the proven implications above.

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Appendix A. Affine–Projection Constants

This appendix records explicit constants for the blockwise short–interval variance bounds used in Theorem 8.1 (variance regime, Part 1) and in Step 12.3 of Part 2.

A.1. **Set-up and Frequency Partition.** Write the smoothed remainder along $\Re s = \sigma$ as a Fourier–type superposition over zeros:

$$H_\sigma(t) = \sum_\rho \frac{e^{(\beta-\sigma)t}}{|\rho|^\alpha} \cos(\gamma t), \qquad \eta_\rho := \beta - \sigma.$$

Differentiating gives

$$H'_\sigma(t) = \sum_\rho \frac{e^{\eta_\rho t}}{|\rho|^\alpha} \big(\eta_\rho \cos(\gamma t) - \gamma \sin(\gamma t)\big).$$

Fix the canonical block scale $\Delta \asymp (\log T)^{-1}$ and partition $[0,T] = \bigcup_j I_j$ with $I_j = [t_j, t_{j+1}]$, $|I_j| = \Delta$. As in Remark 3.2, at the explicit-formula value $\alpha = 1$ the derivative field is taken in truncated form, with frequencies $|\gamma| \le X := \Delta^{-1} \asymp \log T$. In complex notation

$$H'_\sigma(t) = \sum_{|\gamma|\le X} b_\rho \, e^{(\eta_\rho + i\gamma)t}, \qquad |b_\rho| \asymp |\rho|^{-\alpha}|\gamma|, \quad \eta_\rho = \beta - \sigma.$$

The argument below uses only the *global* second moment of this truncated field; its leading term is unconditional, and the sole arithmetic input is the mild ordinate-spacing Hypothesis A.1 (S), far weaker than any large-sieve minimal-gap requirement.

A.2. **Blockwise Affine Projection and a Mean–Variance Lemma.** Let P_j denote the $L^2(I_j)$ projection onto affine functions $\{a+bt\}$ and set $R_j = (\mathrm{Id} - P_j)H_\sigma$. Then $R'_j = H'_\sigma - b_j$, where b_j is the (constant) slope of the affine fit. We have the blockwise stability

$$\int_{I_j} |R'_j(t)|^2 \, dt \;\le\; c_{\text{aff}}^{-1} \int_{I_j} |H'_\sigma(t)|^2 \, dt, \tag{A.1}$$

for an absolute constant $c_{\text{aff}} \in (0,1)$. Indeed, the regression slope $b_j = \frac{12}{\Delta^3}\int_{I_j}(t-\bar{t})\, H_\sigma \, dt = \frac{12}{\Delta^3}\int_{I_j} \kappa_j(t)\, H'_\sigma(t)\, dt$ for a kernel κ_j with $|\kappa_j| \le \Delta^2$, so by Cauchy–Schwarz $\Delta\, b_j^2 \le C \int_{I_j} |H'_\sigma|^2$; combined with $\int_{I_j} |H'_\sigma - b_j|^2 = \int_{I_j} |H'_\sigma|^2 - \Delta\, b_j(2m_j - b_j)$, where $m_j = \Delta^{-1}\int_{I_j} H'_\sigma$ is the block mean of H'_σ and $\Delta\, m_j^2 \le \int_{I_j} |H'_\sigma|^2$, this yields (A.1) with an absolute constant. (We make no use of any specific value of c_{aff}.)

A.3. **Reduction to a global second moment.** Because the blocks tile $[0,T]$, (A.1) gives

$$\sum_j \int_{I_j} |R'_j|^2 \;\le\; c_{\text{aff}}^{-1} \sum_j \int_{I_j} |H'_\sigma|^2 \;=\; c_{\text{aff}}^{-1} \int_0^T |H'_\sigma(t)|^2 \, dt, \tag{A.2}$$

so it suffices to bound the global second moment of the truncated field.

A.4. **Diagonal main term.** Expanding the square,

$$\int_0^T |H'_\sigma|^2\,dt = \sum_{|\gamma|\le X} |b_\rho|^2 \int_0^T e^{2\eta_\rho t}\,dt \;+\; \sum_{\rho\ne\rho'} b_\rho \bar{b}_{\rho'} \int_0^T e^{(\eta_\rho+\eta_{\rho'})t} e^{i(\gamma-\gamma')t}\,dt.$$

In the variance regime $\eta_\rho = \beta-\sigma \le 0$. An on-line zero ($\eta_\rho = 0$) contributes $\int_0^T e^{2\eta_\rho t}dt = T$ to the diagonal; an off-line-left zero ($\eta_\rho < 0$) contributes $\le 1/(2|\eta_\rho|) = O(1)$. Hence the diagonal equals

$$T \sum_{\substack{0<\gamma\le X\\ \beta=\sigma}} |b_\rho|^2\,(1+o(1)) = T \sum_{0<\gamma\le X} \frac{\gamma^2}{|\rho|^{2\alpha}}\,(1+o(1)),$$

the off-line-left zeros giving a lower-order contribution. By Riemann–von Mangoldt, $dN(V) = \frac{1}{2\pi}\log\frac{V}{2\pi}\,dV + O(dV)$, so with $|\rho| \asymp \gamma$,

$$\sum_{0<\gamma\le X} \frac{\gamma^2}{|\rho|^{2\alpha}} \asymp \int_1^X V^{2-2\alpha}\,\frac{\log V}{2\pi}\,dV = \begin{cases} \dfrac{X^{3-2\alpha}\log X}{2\pi(3-2\alpha)}(1+o(1)), & \alpha < \frac{3}{2},\\ O(1), & \alpha > \frac{3}{2}. \end{cases}$$

At the explicit-formula value $\alpha = 1$ this is $\frac{X\log X}{2\pi}(1+o(1)) = \frac{\log T\log\log T}{2\pi}(1+o(1))$ since $X = \Delta^{-1} \asymp \log T$, so the diagonal is

$$T \sum_{0<\gamma\le X} \frac{\gamma^2}{|\rho|^{2\alpha}} = \frac{1}{2\pi}\,T\log T\log\log T\,(1+o(1)). \tag{A.3}$$

This is the source of both the $\log\log T$ factor and the criticality of $\alpha = 1$ (for $\alpha > 1$ the sum converges and the bound improves to $O(T)$).

A.5. **Off-diagonal: negligibility via zero separation.** Using $|e^{(\eta_\rho+\eta_{\rho'})t}| \le 1$ and $\left|\int_0^T e^{i(\gamma-\gamma')t}dt\right| \le 2/|\gamma-\gamma'|$, the off-diagonal is bounded by

$$\Big|\sum_{\rho\ne\rho'}\cdots\Big| \le 2 \sum_{\substack{|\gamma|,|\gamma'|\le X\\ \gamma\ne\gamma'}} \frac{|b_\rho||b_{\rho'}|}{|\gamma-\gamma'|} \le \frac{2}{\delta_{\min}(X)}\Big(\sum_{|\gamma|\le X}|b_\rho|\Big)^2,$$

where $\delta_{\min}(X)$ is the least gap between distinct ordinates with $|\gamma| \le X$. The diagonal carries the full factor T from $\int_0^T$, so the off-diagonal is negligible as soon as $\delta_{\min}(X)^{-1}$ is small relative to T. This is guaranteed by the following mild input, which we state explicitly as it is the *only* arithmetic hypothesis entering the variance bound.

Hypothesis A.1 (Mild ordinate spacing, (S))**.** There is an absolute $c_0 > 0$ such that distinct ordinates $\gamma \ne \gamma'$ of ζ with $|\gamma|, |\gamma'| \le V$ satisfy $|\gamma-\gamma'| \gg V^{-c_0}$; equivalently $\delta_{\min}(V) \gg V^{-c_0}$.

Hypothesis (S) is far weaker than the Riemann Hypothesis or the pair-correlation conjecture: it only forbids distinct ordinates from approaching faster than any fixed polynomial rate, and is consistent with all known zero computations (the closest recorded pairs, e.g. Lehmer pairs, are separated by $\gg 10^{-5}$, vastly larger than the required V^{-c_0} at the relevant heights $V \asymp \log T$). What *can* be proved toward (S), and the precise obstruction to a complete proof, are recorded in §A.6 below; (S) is no easier than the known unconditional small-gap problem for ζ-ordinates, and we isolate it as the single open input. Under (S), $\delta_{\min}(X)^{-1} \ll X^{c_0} = (\log T)^{c_0}$ and, since $\sum_{|\gamma|\le X}|b_\rho| \ll N(X) \ll X^{1+o(1)}$, the off-diagonal is $\ll (\log T)^{c_0+2+o(1)}$ — *polylogarithmic* in T — hence negligible against the diagonal (A.3). The point is structural: the truncation height $X \asymp \log T$ is so small relative to the integration length T that, granting (S), the cross terms cannot compete; no minimal-gap or large-sieve estimate for the (clustered) ordinates is invoked.

A.6. **Toward Hypothesis (S).** We record what can be proved toward (S) and isolate the obstruction.

First, in the variance regime at $\sigma = \frac{1}{2}$ the hypothesis simplifies. If every zero satisfies $\beta \le \frac{1}{2}$, then by the functional equation together with the reflection $\zeta(\bar{s}) = \overline{\zeta(s)}$, each zero $\rho = \beta + i\gamma$ is paired with $1 - \bar{\rho} = (1-\beta) + i\gamma$, also a zero, of real part $1 - \beta \ge \frac{1}{2}$; both can satisfy $\beta \le \frac{1}{2}$ only if $\beta = \frac{1}{2}$. Hence *at $\sigma = \frac{1}{2}$ the variance regime forces all zeros onto the critical line*, and distinct ordinates correspond to distinct zeros with $|\rho - \rho'| = |\gamma - \gamma'|$. For such zeros we have a conditional separation.

Proposition A.2 (Conditional polynomial separation)**.** *Let $\rho_1 = \frac{1}{2} + i\gamma_1,\ \rho_2 = \frac{1}{2} + i\gamma_2$ be distinct on-line zeros with $|\gamma_j| \le V$, set $d = |\gamma_1 - \gamma_2| < 1$ and $s_0 = \frac{1}{2} + i\frac{\gamma_1+\gamma_2}{2}$, and let A be an absolute exponent with $|\zeta(s)| \ll V^A$ on $\{|s - s_0| \le 1\}$ (the functional equation and convexity give polynomial growth in the height on any fixed-width strip, so such an absolute A exists). If there is a point c with $|c - s_0| \le \frac{1}{2}V^{-(A+B)/2}$ and $|\zeta(c)| \ge V^{-B}$, then $d \ge V^{-(A+B)/2}$.*

Proof. Jensen's formula for ζ on the disc $\{|s - c| \le 1\}$ gives

$$\sum_{|z_k - c| \le 1} \log \frac{1}{|z_k - c|} = \frac{1}{2\pi}\int_0^{2\pi} \log\bigl|\zeta(c + e^{i\theta})\bigr|\, d\theta - \log|\zeta(c)| \ \le\ A \log V + B \log V,$$

the sum over zeros z_k of ζ in the disc. Every term is nonnegative, and ρ_1, ρ_2 alone contribute at least $2 \log \frac{1}{d/2 + |c - s_0|}$ (as $|\rho_j - c| \le d/2 + |c - s_0|$). Hence $d/2 + |c - s_0| \ge V^{-(A+B)/2}$, and with $|c - s_0| \le \frac{1}{2}V^{-(A+B)/2}$ this yields $d \ge V^{-(A+B)/2}$. □

Remark A.3 (The obstruction and the status of (S))**.** Proposition A.2 reduces (S) to producing a point c within V^{-c_0} of the cluster at which $|\zeta| \ge V^{-B}$. This is exactly what is *not* available unconditionally: near a cluster of the $O(\log V)$ zeros in a unit disc, Cartan's lemma bounds the small-value set of ζ only up to an exceptional radius of size $O(1)$, not V^{-c_0}, so one cannot in general find a non-small point as close as V^{-c_0} to the cluster. In this sense (S) is no easier than the unconditional small-gap problem for ζ-ordinates — a recognized hard problem. Lehmer's phenomenon shows the gaps can be very small, while random-matrix heuristics predict the smallest gap up to height V to be only polynomially small, comfortably within (S) for a suitable c_0. We stress that (S) is more than what the variance bound needs: (S) makes the off-diagonal *polylogarithmic*, but for it merely to be *negligible* against the diagonal $\asymp T \log T \log\log T$ it suffices that $\delta_{\min}(\log T) \gg T^{-1}(\log T)^{O(1)}$ — i.e. that no two distinct ordinates below height $\log T$ lie within $T^{-1}(\log T)^{O(1)}$ of each other, a vastly weaker requirement, overwhelmingly satisfied since those ordinates are low-height zeros with gaps $\asymp 1/\log\log T$. We therefore leave (S) as the single, mild, numerically unchallenged arithmetic input.

A.7. **Conclusion.** Combining (A.2), (A.3) and the off-diagonal bound (the latter under Hypothesis A.1) gives the global variance bound

$$\sum_j \int_{I_j} |H_\sigma'(t)|^2\, dt \ \le\ C_0(\sigma, \alpha)\, T \log T \log\log T, \qquad C_0 = \tfrac{1}{2\pi} + o(1) \ \ (\alpha = 1), \tag{A.4}$$

and, with (A.1), the form used in Step 12.3 of Part 2:

$$\sum_j \int_{I_j} |R_j'(t)|^2\,dt \;\le\; C_0'(\sigma,\alpha)\,T\log T\log\log T, \qquad C_0' = c_{\mathrm{aff}}^{-1} C_0. \tag{A.5}$$

Remark A.4 (About lower bounds)**.** Only the *upper* variance bound (A.4) is needed for Theorem 8.1 and for the residual control in Part 2. Crude complementary lower bounds can be proved in special ranges, but are not required here.

A.8. **Numerical Verification.** The constants c_{aff}, C_0, and C_0' have been checked in the supplementary notebooks (Part 3) by blockwise evaluation on synthetic signals and Odlyzko windows; see the repository cited in Section 14.

Appendix B. Derivative–Variance (Affine Poincaré) Constant

Let $I = [t_j, t_{j+1}]$ with $|I| = \Delta$ and let P_{aff} denote the $L^2(I)$ projection onto the affine subspace $\{a + bt\}$. For any $r \in H^1(I)$ satisfying the orthogonality conditions

$$P_{\mathrm{aff}} r = 0 \quad\Longleftrightarrow\quad \int_I r(t)\,dt = 0, \qquad \int_I t\,r(t)\,dt = 0,$$

the sharp affine Poincaré (a.k.a. Friedrichs) inequality holds:

$$\int_I |r'(t)|^2\,dt \;\ge\; \frac{4\pi^2}{\Delta^2}\int_I |r(t)|^2\,dt. \tag{B.1}$$

Equivalently,

$$\int_I |r(t)|^2\,dt \;\le\; \frac{\Delta^2}{4\pi^2}\int_I |r'(t)|^2\,dt.$$

Normalization used in Part 2. Writing the bound as

$$\int_I |r'(t)|^2\,dt \;\ge\; \frac{c_{\mathrm{der}}}{\Delta^2}\int_I |r(t)|^2\,dt, \qquad c_{\mathrm{der}} = 4\pi^2, \tag{B.2}$$

matches the scaling adopted in equations (12.1) and (12.6).
Gram matrix and explicit proof. For the basis $\{\phi_1(t), \phi_2(t)\} = \{1, t\}$ on $[0, \Delta]$, the Gram matrix is

$$G_{ij} = \int_0^\Delta \phi_i(t)\phi_j(t)\,dt = \begin{pmatrix} \Delta & \Delta^2/2 \\ \Delta^2/2 & \Delta^3/3 \end{pmatrix}, \qquad \det(G) = \frac{\Delta^4}{12}.$$

Hence

$$G^{-1} = \frac{12}{\Delta^4}\begin{pmatrix} \Delta^3/3 & -\Delta^2/2 \\ -\Delta^2/2 & \Delta \end{pmatrix}.$$

For a function $f \in L^2([0,\Delta])$, the orthogonal projection coefficients (a,b) satisfy

$$\begin{pmatrix} a \\ b \end{pmatrix} = G^{-1}\begin{pmatrix} \int_0^\Delta f \\ \int_0^\Delta tf \end{pmatrix},$$

and the squared projection residual equals

$$\|f\|^2_{L^2(0,\Delta)} - v^\top G^{-1} v, \qquad v = \begin{pmatrix} \int_0^\Delta f \\ \int_0^\Delta tf \end{pmatrix}.$$

This formula underlies Lemma 7.1; together with the Neumann spectral computation below it yields the constant c_{der} in the Rayleigh quotient.

Sharpness. The sharp constant in (B.1) equals the smallest eigenvalue of $-d^2/dt^2$ on $[0,\Delta]$ with Neumann boundary conditions, subject to L^2-orthogonality against span$\{1,t\}$. The corresponding eigenfunction is $\cos(2\pi t/\Delta)$ (the first Neumann mode orthogonal to both 1 and t on $[0,\Delta]$), yielding the sharp constant $c_{\rm der} = 4\pi^2/\Delta^2 \approx 39.5/\Delta^2$. An elementary but weaker bound $12/\Delta^2$ follows directly from the Gram-matrix computation above; since every estimate in Part 2 requires only a *lower* bound on the Rayleigh quotient, either constant is admissible, and we adopt the sharp value $c_{\rm der} = 4\pi^2$.
Discrete grids and numerical check. On the experimental block grids used in Part 3, the discrete least-squares projection onto $\{1,t\}$ and the corresponding finite-difference estimate of $\int_I |r'|^2$ confirm the value $c_{\rm der} = 4\pi^2$ and its Δ^{-2} scaling: the supplementary routine `verify_c_der` reports an empirical minimum of 39.483 for the Rayleigh quotient over random trigonometric residuals, matching $4\pi^2 \approx 39.478$.

Remark. Inequality (B.1) is uniform in the block position t_j and depends only on the block length Δ; no assumptions on r beyond $r \in H^1(I)$ and $P_{\rm aff} r = 0$ are required.

Appendix C. Constant–Extraction Methodology (Optional)

This appendix documents how numerical constants and slopes were extracted from the experiments in Part 3. All scripts and CSVs (`bias_summary.csv`, `bias_blocks.csv`, `variance_table.csv`) are included in the repo and reproduce Figures 1–4 and the tables cited there.

Common settings. Unless otherwise stated, blocks have $\Delta = c_\Delta/\log T$ with $c_\Delta = 2\pi$ (variance) or adaptively chosen $c_\Delta \in [2\pi, 60\pi]$ (bias), and the penalty is $\lambda = c_\lambda(\log T)^{-2}$ with $c_\lambda \in \{1/4, 1, 4\}$. Integrals on each block use the trapezoidal rule with at least 12 samples per block (variance) or 8 (bias). All runs were performed in `R 4.5.1` (double precision).
Data windows. Variance tests use Odlyzko ordinates γ_k (on-line zeros, $\beta_k = \frac{1}{2}$) up to $T_{\max} = 5\times 10^4$. In all experiments, only zeros with $|\gamma_k| \le 200$ are retained (approximately $N(200) \approx 80$ ordinates); this threshold suffices to resolve the harmonic field on the tested interval, and the zero-count cap of 6,000 is never binding. Bias tests augment the on-line set by a synthetic off-line zero at $\beta = \sigma + \eta$, with $\eta \in \{10^{-3}, 5\times 10^{-4}, 10^{-4}\}$.

1) Derivative constant $c_{\rm der}$. We use the affine Poincaré inequality on a block I of length Δ:

$$\int_I |r'(t)|^2\,dt \;\ge\; \frac{4\pi^2}{\Delta^2}\int_I |r(t)|^2\,dt.$$

Thus $c_{\rm der} = 4\pi^2$ in the normalization of Part 2 (see Appendix B). Numerically, we verify the bound by projection onto the affine subspace on a block of length Δ and computing the ratio $\|r'\|^2_{L^2(I)}/\|r\|^2_{L^2(I)}$ over 10^4 randomly phased test signals $\sum_{m=1}^M a_m\cos(\omega_m t + \phi_m)$ with $M \in [1,8]$, $\omega_m\Delta \in [0.1, 10]$. The empirical minimum equals the sharp constant $4\pi^2/\Delta^2 \approx 39.5/\Delta^2$ (empirically 39.483 versus $4\pi^2 = 39.478$; see Appendix B), confirming $c_{\rm der} = 4\pi^2$.

2) Variance constants C_0, C_0'. We bound $\sum_j \int_{I_j} |H_\sigma'(t)|^2 dt \le C_0\, T\log T\log\log T$ as in Appendix A. From `variance_table.csv` (columns F_λ, $T\log T\log\log T$) we regress $Y = \sum_j \int_{I_j} |H_\sigma'|^2$ on $X = T\log T\log\log T$ *through the origin* to obtain $\widehat{C}_0$. Reported intervals are nonparametric percentile bootstrap ($B = 2000$ resamples over the T-grid). Because F_λ grows essentially linearly in $T\log T\log\log T$, the estimate is stable: $\widehat{C}_0 \approx 9.4\times 10^{-3}$ with 95% CI $[9.1\times 10^{-3},\ 1.35\times 10^{-2}]$. (The six-point bootstrap CI is necessarily optimistic but is now narrow, reflecting the monotone growth of F_λ.) The affine residual bound (Equation (A.1)) yields $C_0' = c_{\rm aff}^{-1} C_0$ with $c_{\rm aff} = \frac{1}{4}$.

3) Slope calibration (bias regime). For each (η, T) pair we compute the blockwise SPTB functional for both the baseline (on-line) and biased (off-line zero injected) series. We form the cumulative difference $\Delta F_{\le j} = \sum_{k\le j}(F^{\rm bias}_{\lambda,k} - F^{\rm base}_{\lambda,k})$ and fit

$$\log\bigl(\Delta F_{\le j}\bigr) \;=\; s\, t_j + b + \varepsilon$$

over the last third of blocks (tail window) using Huber regression (`MASS::rlm`, tuning $k = 1.345$) to reduce leverage from early-t transients. The differencing removes the $O(T \log T \log\log T)$ polynomial background, so the extracted slope $\hat{s}$ estimates 2η. Relative errors against the theoretical 2η are recorded in the bias-regime tables of Part 3.

Appendix D. Short Appendix: Dirichlet $L(s, \chi)$ Example

This appendix records the assumptions and standard inputs needed to extend Theorems 8.1 and 12.4 from $\zeta(s)$ to primitive Dirichlet L–functions $L(s, \chi)$ (mod q). The statements and proofs carry over *verbatim*, with the same canonical parameter regime $\Delta \asymp (\log T)^{-1}$ and $\lambda \asymp (\log T)^{-2}$, after replacing the zero–counting function and the Montgomery–Vaughan short–interval bound by their Dirichlet analogues and allowing the variance constant C_0 to depend on q only through $\log(qT)$.

Assumptions and analytic normalization. Let χ be a primitive Dirichlet character modulo $q \ge 1$, and let

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \qquad \Re s > 1.$$

Write the completed L–function in the standard form

$$\Lambda(s, \chi) = \Bigl(\frac{q}{\pi}\Bigr)^{\frac{s+a}{2}} \Gamma\Bigl(\frac{s+a}{2}\Bigr)\, L(s, \chi), \qquad a = \begin{cases} 0, & \chi(-1) = 1, \\ 1, & \chi(-1) = -1, \end{cases}$$

so that $\Lambda(s, \chi) = \varepsilon(\chi)\, \Lambda(1-s, \overline{\chi})$ with $|\varepsilon(\chi)| = 1$. The nontrivial zeros $\rho_\chi = \beta_\chi + i\gamma_\chi$ lie in the critical strip and are symmetric with respect to $1/2$ and the real axis. Along a fixed abscissa $\Re s = \sigma \in [1/2, 1)$ we define the smoothed remainder $H_{\sigma,\chi}(t)$ exactly as in Part 1, with the same truncation and spline block structure. The Dirichlet–series coefficients $\chi(n)$ are bounded and square–summable in the truncated ranges used here.

Zero–counting. Let $N_\chi(T)$ denote the number of zeros $\rho_\chi = \beta_\chi + i\gamma_\chi$ with $0 < \gamma_\chi \le T$. For primitive χ one has the classical formula

$$N_\chi(T) = \frac{T}{\pi} \log\Bigl(\frac{qT}{2\pi}\Bigr) - \frac{T}{\pi} + O(\log(qT)), \tag{D.1}$$

uniformly in $q \ge 1$ and $T \ge 2$. This is the direct analogue of the Riemann–von Mangoldt formula and can be found, for example, in Davenport [9, Ch. 16] or Iwaniec–Kowalski [8].

Short–interval (Montgomery–Vaughan) inequality for Dirichlet polynomials. For Dirichlet polynomials $D(t)=\sum_{n\le N} a_n n^{-it}$ one has the mean–value estimate

$$\int_x^{x+H} \big|D(t)\big|^2\,dt \;\le\; \big(H+O(N)\big)\sum_{n\le N}|a_n|^2,$$

uniformly in $x\in\mathbb{R}$ and $H>0$; see Montgomery–Vaughan [7], or the original short–interval inequalities [2]. In our setting, after freezing slowly varying weights on a block I_j and applying the inequality to $\partial_t H_{\sigma,\chi}$, one obtains the same per–block control as in Appendix A, with the identical $\min\{\Delta,(\gamma^2\Delta)^{-1}\}$ structure. Summing over blocks and using (D.1) yields the global variance bound

$$\sum_j \int_{I_j} |H'_{\sigma,\chi}(t)|^2\,dt \;\le\; C_0(\sigma,\alpha;q)\,T\log(qT)\log\log(qT), \tag{D.2}$$

where $C_0(\sigma,\alpha;q)$ is explicit and depends on q only through $\log(qT)$; for $q=1$ it reduces to the constant $\frac{1}{2\pi}+o(1)$ of Appendix A (the global second-moment diagonal (A.3)).

Transfer of the variance and bias theorems. With (D.2) in place, the proofs of Theorems 8.1 and 12.4 go through without change for $L(s,\chi)$:

- **Variance regime.** If all zeros satisfy $\beta_\chi\le\sigma$, then
$$F_\lambda\big(H_{\sigma,\chi};T,\Delta\big)=O_{\sigma,\alpha}\big(T\log(qT)\log\log(qT)\big),$$
uniformly for $\kappa_1/\log T\le\Delta\le\kappa_2/\log T$ and $c_1(\log T)^{-2}\le\lambda\le c_2(\log T)^{-2}$.
- **Bias/detection regime.** If there is a zero ρ_χ with $\eta:=\beta_\chi-\sigma>0$, then the same assembly as in Part 2 yields
$$F_\lambda\big(H_{\sigma,\chi};T,\Delta\big)\;\gg\;\lambda\,\frac{\eta^2}{|\,\rho_\chi\,|^{2\alpha}}\,\min\Big\{\Delta,\tfrac{1}{|\gamma_\chi|}\Big\}\,\frac{e^{2\eta T}}{\eta\Delta},$$
with implied constants absolute and C_0 replaced by $C_0(\sigma,\alpha;q)$.

Remark on possible exceptional (Siegel) zeros. For real characters χ there may exist a *Siegel zero* $\beta_\chi=1-\delta(q)$ with $\delta(q)\ll(\log q)^{-A}$. Such an off–line zero (when $\sigma<\beta_\chi$) is *exactly* the kind of input detected by the bias theorem: the exponential rate $2(\beta_\chi-\sigma)$ appears unchanged. Our arguments do not assume the nonexistence of such zeros; they merely show that their presence forces exponential growth of F_λ.

Conclusion. All statements of Parts 1–2 extend to $L(s,\chi)$ with the single substitution $\log T\rightsquigarrow\log(qT)$ in the variance scale and with a possibly different variance constant $C_0(\sigma,\alpha;q)$; the detection direction and exponential rate $2(\beta_\chi-\sigma)$ are unchanged.

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